

Quantifying the prevalence and impact of overreaching causal claims in social science

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Across the social sciences, many studies use cross-sectional designs that reveal associations but are generally unable to support direct causal claims, yet authors of such articles may make or imply causal claims anyway. Here, to examine the prevalence of such ‘overreaching’ causal language, we analysed 194,631 cross-sectional articles using large language models. Over the period 1980–2024, an average of 46% of articles contained causal language in their titles or abstracts, where the annual rate has risen almost threefold since 2000 from 20% to 60%. To examine the effects of such language, we conducted a human-subjects experiment ($N = 1,105$), finding that readers frequently indicate abstracts with this phrasing provide causal evidence but that methodological labels ($\beta = -0.4$, 95% confidence interval -0.56 to -0.19) and associational wording ($\beta = -0.3$, 95% confidence interval -0.43 to -0.07) reduce this tendency. Experiments with five LLMs revealed that model summaries of these articles ($N = 100$ each) can amplify causal overstatement, removing hedges and introducing causal claims where articles used strictly associational phrasing; however, prompting caution diminishes this pattern.

Reliable and trustworthy science communication requires measured claims that accurately reflect empirical evidence^{1,2}. In academic publications, this ideal translates into reporting research questions, methods, results and implications in a way that avoids distortion and misleading readers^{3,4}. Yet scientific articles are not purely objective accounts of facts. Researchers retain considerable discretion in how they operationalize constructs and rhetorically frame their findings^{5–7}, providing flexibility that also opens the door to misinterpretation. An important case of what we call ‘Narrative License’ occurs when scholars claim or imply that correlational findings are causal without conducting the appropriate tests⁸. In such cases, readers seeking evidence-based solutions⁹ may find themselves persuaded of causal claims that are not based on causal evidence.

Conflating correlation with causation can yield serious real-world consequences when such claims inform policy, medicine, or communication. For example, in political science, influential work argued that civil wars were driven not by poverty or ethnic grievances but by structural opportunities for insurgency^{10,11}. Although correlational, these findings were widely characterized as causal, including in *The*

New York Times, and were credited with shaping World Bank policy. Yet, subsequent work found that the models in question performed poorly at predicting outcomes¹², implying that interventions based on these models could misallocate resources. Similar problems have arisen in public health: observational analyses from the Nurses’ Health Study linked postmenopausal estrogen therapy to reduced cardiovascular risk¹³, but randomized trials later showed the opposite¹⁴, with benefits limited to younger women¹⁵. Uncritical application of the original causal claims could result in interventions that do more harm than good and erode public trust. Similar overclaiming also appeared in media-effects research linking violent media to aggression^{16–18}, with the relationship between the two remaining empirically contested^{19–22}. Such cases illustrate how correlational findings are often interpreted as causal^{23,24}, sometimes amplified by scholarly language.

Concerns about conflating correlation with causation are longstanding in the social sciences; however, the prevalence of such conflation has been difficult to establish. In contrast to clearly spurious claims (for example, featured in humorous compilations, <https://tylervigen.com/spurious-correlations>), most scholarly articles test theoretically

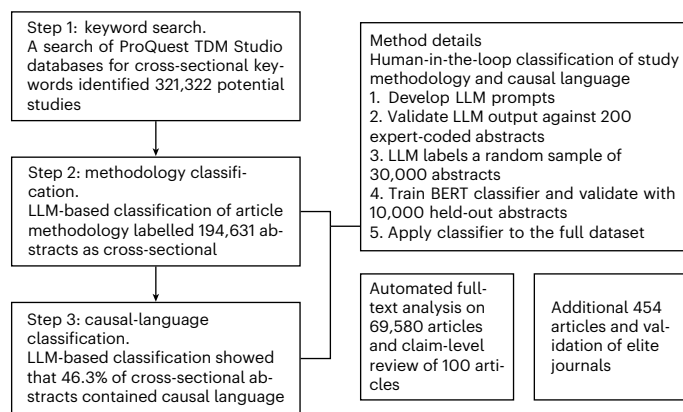


Fig. 1 | Overview of the three-step LLM-based workflow used to identify cross-sectional studies and detect causal language across five ProQuest databases. Step 1: we queried databases in the ProQuest walled-garden, data-mining platform TDM Studio using keywords associated with cross-sectional methods and exported 40,000 titles/abstracts to develop and evaluate models. Step 2: to verify that articles were truly cross-sectional, we designed a GPT-4o prompt and validated it against 200 expert-coded articles. The validated prompt was applied to all retrieved abstracts, and a fine-tuned BERT classifier (trained on 30,000/evaluated on 10,000) reproduced GPT and expert judgements. Applying this classifier to all keyword-identified article abstracts yielded 194,631 cross-sectional studies. Step 3: using a parallel workflow, we coded causal language in titles and abstracts. A fine-tuned BERT classifier again matched GPT and expert labels, allowing classification of all cross-sectional articles and revealing that 46.3% used causal language, a notably high rate given the limited identification leverage of such designs. We additionally analysed full texts when available and supplemented the dataset with articles from general science journals not included in ProQuest.

plausible hypotheses and use empirical designs that can support causal inference under at least some conditions. The resulting ambiguity makes it difficult to systematically evaluate when causal claims overreach, especially at the scale of whole literatures.

To simplify the problem, we focus on empirical articles using cross-sectional, non-experimental designs, excluding those with longitudinal components. By definition, such designs cannot establish the temporal ordering of cause and effect (that is, Bradford Hill's temporality criterion^{25,26}), and non-experimental data, lacking randomization, is generally susceptible to confounding variables and reverse causality. We recognize, of course, that quasi-experimental methods such as instrumental variables and regression discontinuity designs allow for causal conclusions to be drawn from observational snapshots when investigators make strong, explicit assumptions and articulate a credible identification strategy^{27–30}. We therefore exclude designs of this sort, such that the remaining corpus of studies rely on purely cross-sectional variation, rendering their causal claims sensitive to untestable identification assumptions. We further note that associational evidence, even if not causal on its own, could reasonably be interpreted as adding support to causal claims derived from theory, models or other empirical evidence. For this reason, we designed a classifier to code causal claims as overreaching only when they were made about the data under consideration, not targeting claims presented as correlational, though 'aligning' with causal theories.

Even with these restrictions, studying the prevalence of causal claims in published articles at scale poses challenges. Authors are often aware of methodological limits and may use associational language when describing results, yet still imply causality by framing their work as testing causal hypotheses or by endorsing implications that only follow under causal interpretation. We also wish to examine differences in prevalence across several social science disciplines, which differ in topics, terminology and article structure, thereby complicating the

analysis in ways traditional text analysis methods^{31–35} often cannot handle reliably.

To address these difficulties, we develop methods that leverage the contextual understanding of large language models (LLMs). LLMs have been shown to perform well on diverse text-analysis tasks³⁶, often matching or exceeding state-of-the-art models and crowd workers^{37,38}, even with zero-shot prompts³⁹. Nonetheless, expert validation remains essential^{40,41}, especially as LLMs are increasingly integrated into research systems^{42–44}.

Accordingly, we developed a three-step LLM-based workflow with human-in-the-loop validation to analyse social science articles, primarily focusing on titles and abstracts, which are likely to receive disproportionate reader attention. As illustrated schematically in Fig. 1, we first queried five ProQuest databases using keywords commonly associated with cross-sectional methodologies. Second, we exported a sample of titles and abstracts and constructed an LLM prompt to classify study methodology as cross-sectional or not (Extended Data Box 1). The resulting labels closely matched expert human judgements and were reproducible using a fine-tuned BERT-based classifier. Applying this classifier to the full dataset, we identified 194,631 studies that relied exclusively on a cross-sectional design (that is, did not contain a mixture of cross-sectional and longitudinal or (quasi-)experimental designs). Third, we used a parallel procedure to code causal language in titles and abstracts (see Table 1 for example claims; see Extended Data Box 2 for the prompt). A fine-tuned BERT classifier again replicated GPT- and expert-generated labels, enabling us to classify all cross-sectional articles for the presence of causal language and to examine temporal trends and variation across disciplines and journals. Further, we analysed the full texts of a subset of articles to explore whether this increased contextual information presented higher or lower rates of causal claims, supplementing the dataset with additional full texts and general science journal articles not represented in ProQuest (see the Methods for more details of all stages of our pipeline and robustness checks).

Beyond documenting the prevalence of causal language, we also examined its impact on readers. People often infer causality from associations^{45,46}, suggesting that overclaiming may stem as much from a cognitive 'default towards causality' as from career or institutional pressures. To test how causal phrasing shapes interpretation, we ran experiments in which human participants read an abstract drawn from a stratified random sample of articles in elite journals within our dataset. Each abstract appeared either in its original form, containing causal language, or in a version rewritten to use purely associational phrasing. After reading, participants judged whether the study provided causal evidence, summarized findings (allowing us to code causal language) and evaluated hypothetical interventions that would only work if effects were causal. Following prior work^{8,47}, we also tested a brief methodological label and an artificial intelligence, 'AI', warning that highlighted overclaiming as potential tools to improve reader inference.

Finally, given the increasing integration of LLMs into scientific workflows and building on research that analyses LLM outputs^{48,49}, we conducted a parallel experiment with GPT-4.1 to examine how its summaries shift under the same interventions. Using LLMs could exacerbate potential human misperception if, say, model summaries tend to present correlational evidence as causal, potentially generating summaries that are more misleading than the originals. To test this possibility, we further examined model summaries of a larger set of articles with varying types of causal language and evaluated how prompting strategies influenced different models' output, expanding the analysis across five leading LLMs.

Through this combination of large-scale descriptive analysis and controlled experimentation, we aim to contribute to meta-knowledge⁵⁰ in three ways: first, by clarifying how often causal language appears in cross-sectional social science research; second,

Table 1 | Ten randomly selected examples of causal claims in cross-sectional titles and abstracts, as coded by our classifiers

GOID	Quote
521097004	"...an unfavorable effect of 'time restriction' on well-being is expected...[we show that] the time restriction factor adds to the degree of exhaustion and the work-nonwork conflict, while time autonomy diminishes these outcome variables."
194739887	"Role of multihospital system membership in electronic medical record adoption...Multiple regression analysis was used to examine the impact of multihospital system affiliation on EMR level of adoption."
2847421536	"...we show that being in a dual-career household increases one's willingness and lowers the perceived risk of leaving their job and joining a startup venture."
1826809431	"[Our findings about the] moderation of self-regulatory efficacy suggests self-regulation minimises individual engagement in deviant acts. Thus, human resources managers in Nigerian universities should consider self-regulatory efficacy as a selection criterion when hiring academicians."
1966300327	"While prevalence studies have examined the incidence of problem gambling in other age groups, little attention has been paid to the impact of casino gambling on older adults. This study investigated the prevalence of casino gambling as a social activity for active senior citizens."
2135260445	"This study considers the effect of leadership and managerial constructs on lecturers' commitment to the newly implemented honours programmes in a Dutch University...visionary leadership and the perceived discussion culture on excellence are of large influence on lecturers' behaviour."
1647802301	"We investigate both the role of gender and feminism in [Friend's With Benefits] relationships at a United States college, and ask whether identification with feminist ideology impacts students' motivations and assessments of their relationships."
2627195225	"The disproportionate rates of police surveillance and encounters in many communities in the US may be contributing to inequities in health and violence."
1895058836	"It was found that only [stakeholder engagement/empowerment] has a direct positive impact on [project success]."
220966357	"The results show that policy ownership has a substantial, positive impact on service usage and costs, particularly for beneficiaries in fair or poor health. The greatest impact was found for policies that provide first-dollar coverage."

IDs correspond to the unique GOID identifier from ProQuest, which can also be used to find article citation information in the Supplementary Information.

by assessing its practical consequences for audience interpretation and LLM summaries; and third, by testing potential interventions to improve communication.

Results

Prevalence of causal claims in academic articles

Figure 2 shows that from 1980 to 2024 our classifiers coded 46.3% of the 194,631 abstracts as making causal claims about the paper's empirical results. As we have just argued, cross-sectional designs are poorly suited to support such claims; we therefore describe these claims as 'overreaching' or 'inappropriate' and note that their near-half prevalence is strikingly high. Figure 2 also shows a sharp increase since 2000, rising from a relatively stable 20% to over 60% by 2024. As an additional baseline, we also compared this rate with the prevalence of causal claims in a separate sample of (quasi-)experimental studies, finding that the latter is higher; however, this discrepancy has diminished over time ('Comparison to experimental studies' section in the Supplementary Information). Across five disciplines the prevalence increased over time, but by 2024, business was highest (84%) and sociology lowest (46%), while economics (67%), psychology (55%) and political science (53%) sit nearer the overall average. Extended Data Fig. 1 shows that

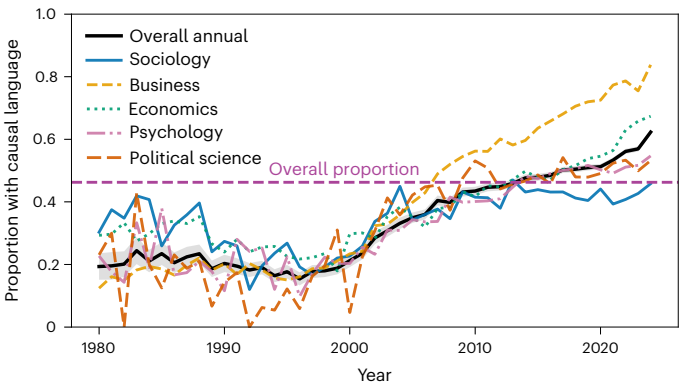


Fig. 2 | The annual proportion of cross-sectional articles that contain causal language across all articles (black) and by discipline (colours). Overall, the proportion remained stable at roughly 20% from 1980 to 2000, then increased rapidly tripling by the 2020s. This increase appears across all disciplines, with notable recent variation: business shows the highest rates, whereas sociology shows the lowest. The grey band represents the annual 95% Wald CI for all articles. The horizontal dashed line marks the pooled overall proportion across all articles (46.3%, $N = 194,631$), which is disproportionately influenced by the higher volume of articles in later years.

article counts grew rapidly from 1980 onwards, weighting the overall average towards recent years, and Extended Data Fig. 2 displays disciplinary counts.

We next examined variation in causal claims across journals with different prestige and impact. For this analysis, we focused on the 127,598 cross-sectional articles for which we could link SCImago Journal Rank (SJR) data. Interestingly, causal language was more prevalent among higher-impact journals (Pearson's $r(1,550) = 0.22$, 95% confidence interval (CI) 0.176 to 0.271, $P < 0.001$; among journals with ten or more articles). As Fig. 3a shows, among journals above the median impact factor >3.03 , 54.4% of abstracts contained causal language, compared with 43.3% in below-median journals. A multilevel logistic regression of causal language on impact-factor quintile, with year fixed effects and journal-clustered standard errors, showed that this association persists within year ('Within-year impact-factor differences' section in the Supplementary Information). Exploratory analyses that separate claims into different types show that increases over time and with impact factor are driven mainly by an increase in articles with stronger direct and implied claims rather than by more careful, conditional phrasing (Extended Data Figs. 3 and 4; 'Results by causal type' section in the Supplementary Information).

To further examine the relationship between journal prestige and causal language, we analysed articles from five leading journals in each of four disciplines (economics, political science, sociology and general social science; 20 journals total) and added a newer journal, the *Journal of Quantitative Description: Digital Media*, that explicitly disallows causal claims. For these journals, we manually validated both methodological classifications and causal-language labels ('Validating elite journals' section in the Supplementary Information) and then plotted the proportion of abstracts containing causal language by journal group (Fig. 3b). Across disciplines, elite journals showed a causal-language rate of 43.2% slightly lower ($Z = -2.1$, $P = 0.039$, Wilson 95% CI 0.40 to 0.46) than the proportion observed in the broader dataset (46.3%). As expected, the *Journal of Quantitative Description: Digital Media* showed substantially lower rates of causal language (two-proportion $Z = -6.73$, $P < 0.001$, Δ proportion = -0.33 , 95% CI -0.40 to -0.27), although causal language was still present in 12.9% of its articles (Wilson 95% CI 0.08 to 0.21).

Although claims made in the title and abstract are, by design, the most prominent claims made in a paper, the full text also offers authors many opportunities to assert claims that may reinforce, moderate

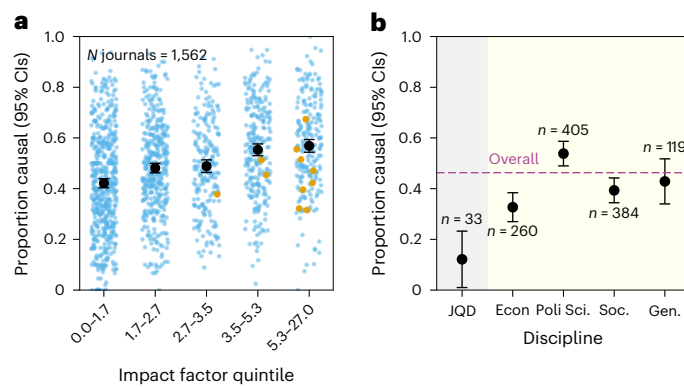


Fig. 3 | Proportion of cross-sectional articles containing causal claims, stratified by journal impact factor and discipline. **a**, Proportion of cross-sectional abstracts with causal language by journal impact factor quintile. Blue dots represent proportions for individual journals with ten or more articles (N journals = 1,562). Black dots represent the mean proportion across journals in each quintile. The error bar reflects a 95% CI of the mean. Orange dots represent top journals in **b**. This plot shows that causal language is more common in higher-impact journals. **b**, The proportion of cross-sectional abstracts with causal language from five elite journals in four different disciplines (economics (Econ), political science (Poli Sci.), sociology (Soc.) and general science (Gen.), $N = 20$ total) and a new journal dedicated to using only descriptive language, the *Journal of Quantitative Description: Digital Media* (JQD), with 95% Wald CIs. The dashed line represents the proportion across all abstracts from all journals. Elite journals across disciplines exhibit substantial rates of causal language in cross-sectional articles, similar to the overall proportion. The number of articles from journals for each discipline is displayed in the figure.

or even contradict their message. Using the same causal-language classifier, we analysed full-text snippets from 69,580 cross-sectional articles available via TDM Studio ('Full text analysis' section in the Supplementary Information). As shown in Supplementary Fig. 3, 81.8% contained at least one causal snippet, well above title/abstract rates, suggesting authors often adopt causal framing in the main text even when abstracts are cautious. However, full texts contain many more total claims than abstracts, so the impact of these causal claims may be tempered by other associational statements.

To test for this possibility, we conducted a second analysis in which we searched for specific claims in the full texts of a smaller subsample of articles ($N = 100$) that closely matched claims made within the abstract (see 'Full text analysis' section in the Supplementary Information for details). On the one hand, the majority of matched claims were associational or descriptive ($N = 2,287$; 73.0%), followed by strong, unhedged causal ($N = 487$; 15.5%) and more careful, conditional causal claims ($N = 360$; 11.5%). Thus, only 26.8% of matched claims were overreaching at the claim level, significantly lower than the prevalence of overclaiming measured at the abstract level ($z = -22.2$, $P < 0.001$, Δ proportion = -0.19 , 95% CI -0.21 to -0.18). On the other hand, consistent with our first analysis, articles frequently included at least one unhedged causal claim even when the title and abstract did not use causal language. As shown in Fig. 4a, 100% of articles that contained an unhedged causal claim in the title or abstract also contained at least one matched causal claim in the full text. Moreover, 72% of articles with conditional or implied claims and 44% with only descriptive claims in the title/abstract had at least one unhedged causal match in the full text. Figure 4b shows the average number of matched claims identified in each manuscript section, with the proportion classified as unhedged causal. At the paper level, articles most commonly contained at least one unhedged causal claim in the introduction (60.0% of articles, 95% CI 50.1% to 69.9%) or discussion (55.1%, 95% CI 45.3% to 64.9%), followed by the results (42.7%, 95% CI 32.4% to 53.0%) and only rarely in the methods (6.1%, 95% CI 0.3% to 11.8%).

Impact on human interpretations and LLM summaries

Our descriptive analysis revealed the widespread prevalence of causal claims in cross-sectional work but also raises the question of how this narrative overreach impacts reader inferences—and, if so, whether there are simple interventions to improve clarity. To answer these questions, we conducted an experiment to test how overreaching causal language in abstracts influences reader judgments. College-educated US adults ($N = 1,105$) were randomly assigned to one of four conditions: (1) the original abstract containing a causal claim, (2) a rewritten version using strictly associational language, (3) the original abstract with a methodological note stating that the study was cross-sectional and therefore cannot on its own establish causality or (4) the same note accompanied by 'AI' feedback highlighting methodological and interpretive issues (Extended Data Fig. 5). Each participant read one of 28 abstracts sampled from elite journals, wrote a brief summary and rated its causal implications (full preregistration, materials and analytic models are available in the Methods and the Supplementary Information). The human-subjects experiment was preregistered, and we report one-sided P values with Holm–Bonferroni corrections applied within each model family. A second experiment presented the same stimuli to GPT-4.1 to assess the influence of language on model output. We emphasize that these two experiments answer different questions: whereas the human experiment concerns how readers interpret causal claims, the experiment with GPT explores how AI-generated summaries might distort research findings.

Human participants

Overall, Fig. 5 (orange symbols) shows that human respondents have a strong latent tendency to make causal inferences regardless of the evidence and that this tendency is resistant to corrective interventions but can be partially rectified for at least some outcomes. Extended Data Table 1 reports inferential statistics, model estimates and CIs for all analyses summarized below.

First, Fig. 5a shows that in the original abstract condition, participants tended to agree that the studies provided causal evidence. In comparisons, participants who received a rewritten abstract with purely associational language, a methodological note or 'AI' feedback showed lower agreement, with medium effect sizes (standardized β range from -0.25 to -0.55 ; $P \leq 0.003$; Extended Data Table 2, top). Thus, removing or flagging causal language influenced reader interpretations, yet participants frequently inferred causal evidence regardless of condition, even when causal claims were absent or explicitly denied.

Second, Fig. 5b presents responses to a question about a hypothetical manipulation of the independent variable, implicitly invoking causality but not asking about the paper's evidence directly. Here, human respondents were even more likely to agree that the proposed relationship is causal and even less responsive to the interventions, with no evidence of significant difference observed across conditions ($\beta = -0.12$ to -0.18 ; $P = 0.073$ to 0.092 ; Extended Data Table 1, middle). These results could arise either because humans are more likely to make causal inferences when asked about them indirectly or because responses may reflect prior beliefs rather than paper-based evidence. To distinguish these, we included a 'No Info' control condition in which participants also reported beliefs before reading the paper. We observed no evidence that responses differed from those in the original abstract condition ($\beta = -0.07$, $P = 0.471$). However, participants in the 'No Info' condition rated their beliefs both before and after reading the abstract, and a paired-samples t -test indicated a small increase in implied causal beliefs ($t(217) = 2.17$, $P = 0.031$, mean change = 0.15 , 95% CI 0.014 , 0.293). Taken together, these results suggest that exposure to the abstract may have shifted beliefs modestly, but that the magnitude of this shift was small relative to participants' pre-existing beliefs.

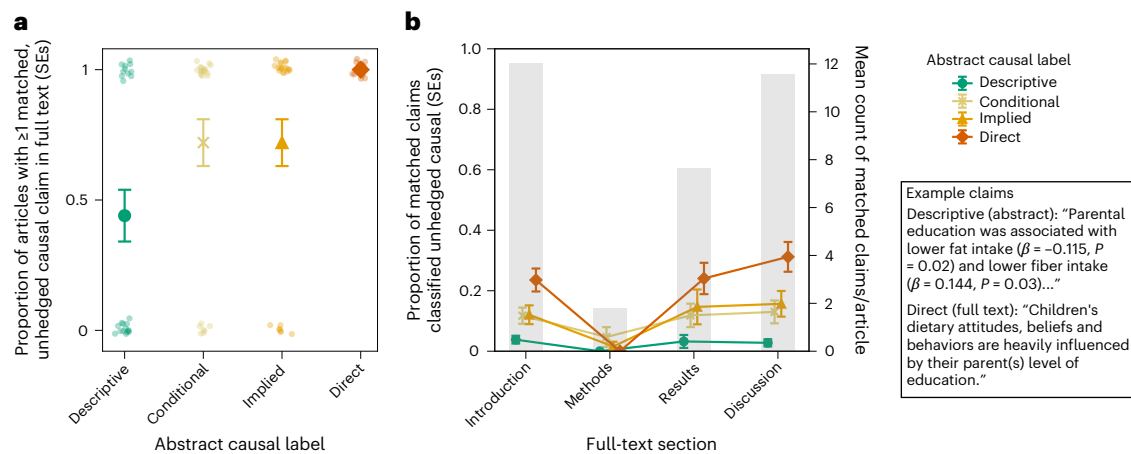


Fig. 4 | Causal claims present in the full text, grouped by article abstract-level classifications. **a**, Proportion of articles with at least one unhedged causal claim in the full text that matched a claim made about the article's findings within the abstract. Error bars represent the standard error of the proportion (SEs). Even articles with abstracts that contain only conditional or descriptive claims frequently include unhedged causal claims in the full text that otherwise match the more carefully worded claim from the abstract. For this analysis, we manually identified a stratified sample of 100 articles, comprising 25 from each of the four abstract claim categories. **b**, Lines: proportion of matched claims labelled unhedged causal by section and abstract-level causal classification. Bars: mean

number of matched claims in each section. Across abstract-level categories, most matched claims in the full text remain associational, and the unhedged causal claims that do appear are concentrated in the introduction and discussion, with far fewer in the methods. This lower proportion reflects both the smaller number of matched claims ($N_{\text{intro}} = 1,150$; $N_{\text{method}} = 168$; $N_{\text{results}} = 720$; $N_{\text{discussion}} = 1,099$) and their lower likelihood of being causal (proportion of claims unhedged causal: intro, 17%; methods, 4%; results, 14%; discussion, 17%; $\chi^2(6) = 81.0$, $P < 0.001$). Right: an example of a matched abstract-full-text claim pair is shown. Error bars represent standard errors. Error bars for **a** and **b** represent normal approximated standard errors.

Finally, Fig. 5c shows how likely participants were to use causal language when summarizing the paper's findings (see Fig. 5 for example summaries). Participant summaries in the original-study condition were very likely to contain causal language. The 'rewrite' (odds ratio (OR) of 0.38, $P < 0.001$) and AI (OR of 0.60, $P = 0.022$) conditions showed significant reductions in this tendency, and although the note condition was in the expected direction, it did not show evidence of a significant difference (OR of 0.73, $P = 0.078$). Across all conditions, causal language was common in participants' summaries (original, 71.3%; rewrite, 49.6%; note, 64.2%; AI, 59.4%) again suggesting that the tendency to impute causality to observed associations is strong by default and challenging to correct (Extended Data Table 1, bottom).

First experiment on GPT-4.1

In addition to our human-subjects experiment, we conducted a second experiment in which we provided the same experimental stimuli to GPT-4.1. Extended Data Table 2 reports full model results, which we summarize here. Overall, Fig. 5 (blue symbols) shows that the model frequently described correlational findings as causal. Specifically, Fig. 5a shows that in the original condition GPT output was unlikely to report that the study provided causal evidence and showed large reductions in the rewrite, methodological note and AI feedback conditions (β range of -1.23 to -1.75 , all $P < 0.001$). In the AI condition, for which the effect was largest, model output showed strong disagreement that the study provided causal evidence (mean of 1.41, 95% CI 1.24 to 1.58). For the implicit causal inference outcome, Fig. 5b shows that GPT output was slightly more likely to agree with the causal interpretation after receiving the original abstract than in the 'No Info' condition, whereas no such increase was observed when the abstract was rewritten using associational language. As with the explicit question in Fig. 5a, however, the rewrite, note and AI feedback interventions all significantly reduced the likelihood to agree with causal interpretation (β range of -0.60 to -1.79 , all $P < 0.001$). Finally, Fig. 5c shows that model-generated summaries based on the original abstracts almost always included causal claims (99.3%), but that once again, all three interventions showed significantly lower likelihoods (rewrite, 89.8%; methodological note, 41.7%; AI feedback, 30.3%; all $P < 0.001$). Notably,

the note and AI treatments had quite large effects, but no intervention fully eliminated causal framing.

Second experiment on LLMs

The experiment just discussed had four important limitations: it relied on a single prompt, used a single model, did not account for the type of causal claim in the abstract and relied only on abstracts rather than full texts. To address these limitations, we conducted a second experiment that used four summarization prompts: a basic summarization, similar to the previous experiment; a simplifying summary, in which the models were asked to 'use plain, everyday words and short sentences' at an eighth-grade reading level; a summary with a specific focus on practical impact; and a 'careful' summary that focused on methodological limitations ('Second experiment with LLMs: prompts' section in the Supplementary Information). For our outcome, we used a new dedicated classifier tailored to these summaries. In addition, we sampled articles stratified by the three types of causal claims—direct, implied and conditional—as well as descriptive (that is, no causality claimed), and we tested five different models (GPT-4.1, GPT-5.1, GPT-5.4, DeepSeek Chat-V3.2 and Anthropic's Claude Sonnet 4.6). Finally, we ran a separate condition in which each model was given the full text of the article rather than the title and abstract alone. Stuart–Maxwell tests revealed no evidence of significant differences in causal language between full-text and abstract-based summaries for any prompt or model (Supplementary Table 7); thus, we present results for abstracts only here.

Figure 6a shows that, across this more diverse sample of articles, LLM-generated summaries contained unhedged causal language at substantially higher rates than the abstracts themselves overall, with notable variation across prompts (see Supplementary Table 8 for full 2×2 tables and Extended Data Table 3 for corresponding statistics). Under the basic summarization prompt, GPT-4.1 and DeepSeek Chat-V3.2 produced significantly more causal claims than the abstracts, whereas the remaining models did not show evidence of a significant difference from the abstracts after correcting for multiple comparisons. The simplifying and practical prompts also yielded higher rates of causal claims, this time across all models (see Extended Data Table 3

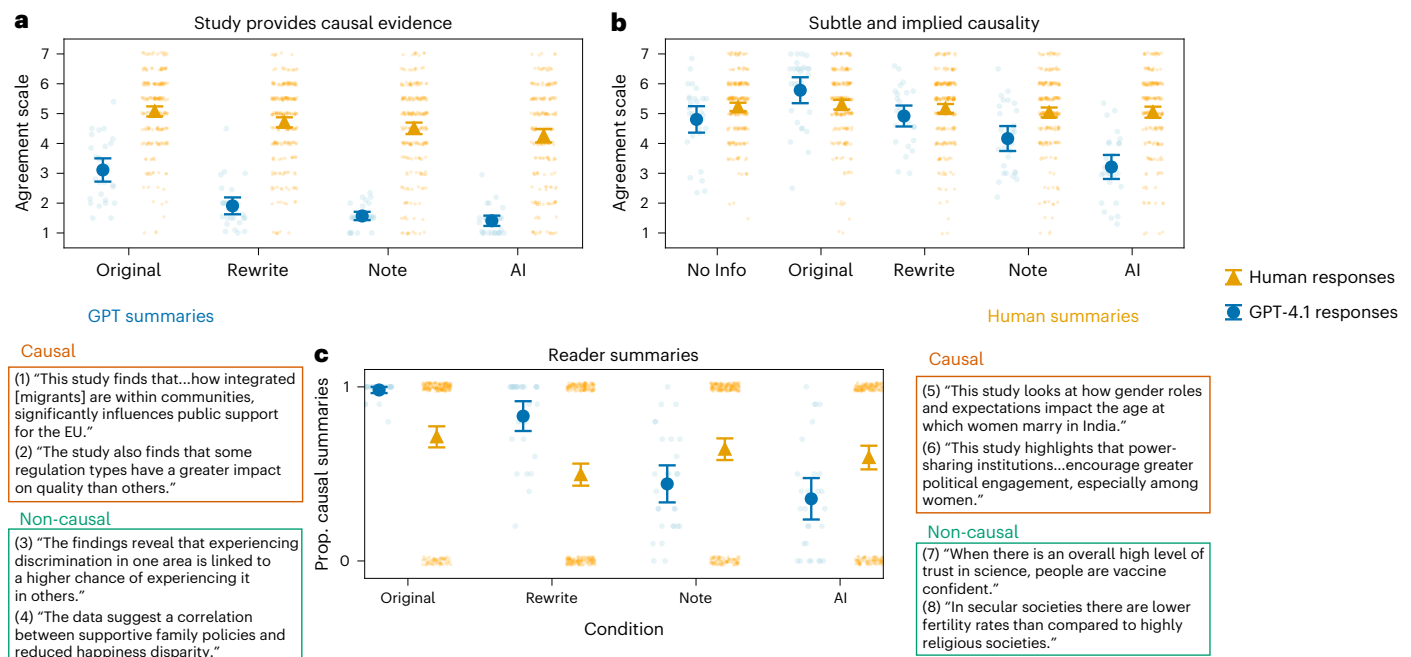


Fig. 5 | Causal beliefs indicated by human participants and in GPT-4.1

summaries across experimental conditions. For this experiment, 1,105 human participants were randomly assigned to the different conditions, where they read one of 28 article abstracts: *N* control = 218, original = 216, rewrite = 240, note = 229 and AI = 202. GPT-4.1 produced 10 responses for each of the 28 abstracts. **a**, Human participants were more likely to agree that the study established a causal relationship when reading the original abstracts compared with model outputs; both the rewrite and the two interventions significantly reduced agreement for human participants and in GPT-4.1 output. Thicker points and error bars show means and 95% CIs, for both **a** and **b**. **b**, GPT-4.1 output showed similar reductions

on subtler items implying causality, and while human responses followed the same direction, they did not show evidence of a significant difference across conditions. **c**, Free-text research summaries from humans and GPT-4.1 frequently contained causal claims in the original condition; the interventions reduced this tendency, though the summaries continued to contain causal claims across conditions. GPT-4.1 produced ten summaries per paper, so the model's causal-language measure ranges from 0 to 1, reflecting proportions (Prop.) rather than the binary judgements used for individual human responses. Thicker points and error bars show proportions with 95% CIs. Example snippets from summaries with and without causal language are shown alongside the plot.

for *P* values). By contrast, the careful prompt (mean of 5.0%) reduced causal language across all models except GPT-4.1.

A multilevel logistic regression confirmed that these differences across prompts and models were statistically meaningful when accounting for repeated measures by article (Supplementary Table 9). It further showed that abstracts containing any form of causal language (conditional, implied or direct) were more likely to yield summaries with unhedged causal claims relative to purely descriptive abstracts, with predicted probabilities increasing with the strength of the original causal language. Figure 6b illustrates how causal claims shift from abstracts to summaries across models for each prompt condition.

In short, Fig. 6 largely replicates and extends the main LLM insights of Fig. 5: language models appear prone to produce summaries of empirical work that overstate causality even when the evidence is purely associational, but they also are responsive to corrective interventions, including a 'careful' prompt which reduced causal overstatement relative to abstracts.

Discussion

In this study, we have demonstrated that overreaching causal language is pervasive across the social sciences. Restricting our analysis to non-experimental studies that rely exclusively on cross-sectional data, we find that nearly half (46.3%) of such articles published since 1980 contained causal claims in either their titles or abstracts. Even this large number understates the prevalence in recent years, which is over 60%, reflecting a threefold increase since 2000—a trend that occurred across all disciplines, albeit to different extents. Interestingly, these results applied equally to the most prestigious and selective journals and somewhat more so to high-impact journals in general. Moreover,

full-text analyses showed that even when abstracts used cautious, associational language, corresponding articles often included direct causal statements elsewhere. Together, these results, which align with other evidence of causal overclaiming in psychology⁵¹ and biomedical science³⁵, indicate that rhetorical overreach is not limited to isolated cases or to low-status outlets but reflects a systemic feature of how correlational findings are communicated.

In addition to studying the prevalence of inappropriate causal claims, we also studied their effects on humans and language model summaries in a series of three experiments in which we also tested various interventions to improve reader inferences. For humans, we found that our interventions somewhat reduced the share of respondents who agreed that the paper in question provided causal evidence but that respondents continued to perceive causality at surprisingly high rates in all conditions. Even when abstracts were rewritten in strictly associational terms or corrective notes were added to the text, participants often reintroduced causal phrasing in their summaries. These findings suggest that causal interpretation is deeply ingrained in humans and is difficult to eliminate even with more careful phrasing. Our experiments revealed that LLMs frequently introduced causal language when summarizing study findings, under basic, simplifying and practical-oriented prompting strategies. This mirrors recent evidence that LLMs overclaim when summarizing scientific findings, for instance, by overgeneralizing⁵². However, LLMs proved substantially responsive to intervention. Simple textual augmentations, study contribution labels and a 'careful' prompt requesting caution in summarization all substantially reduced the presence of causal language. These results not only raise concerns about the integration of LLMs into research and communication workflows but also suggest that simple prompting modifications can ameliorate potential misinterpretations within model summaries.

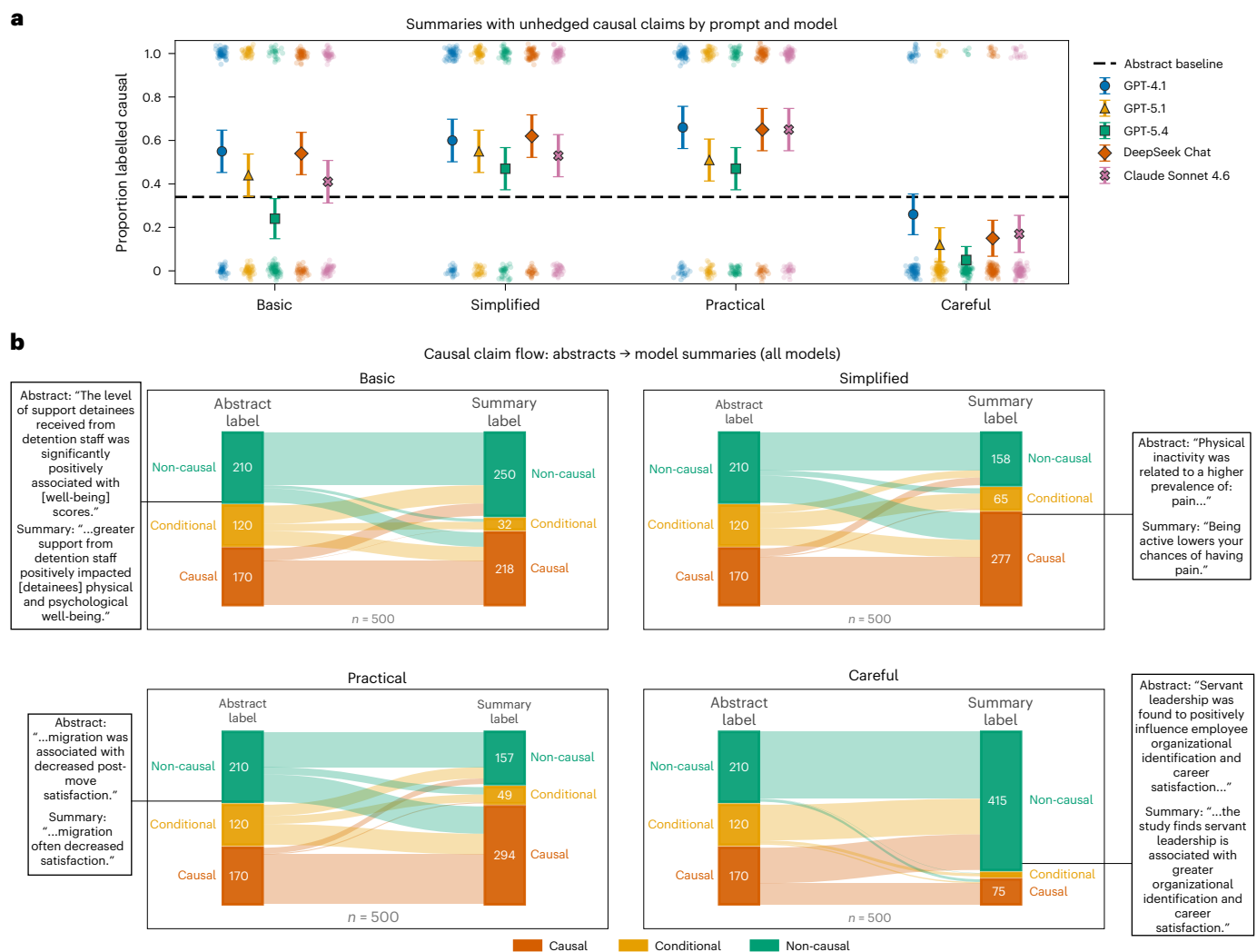


Fig. 6 | Causal language in LLM-generated article summaries across prompting strategies and models. a, The proportion of summaries from each of the five models tested that contain unhedged causal statements, with the horizontal line representing the baseline rate in abstracts. Error bars represent 95% Wilson CIs. Each model generated a summary for 100 articles under each of the four prompts, yielding a total $N = 2,000$. Simplifying and practical prompts produced significantly higher causal claim rates across all models; basic prompts

also led to higher causal claim rates, except for GPT-5.4. A careful prompt reduced causal language across models. **b**, How claim types shift from abstracts to summaries are tracked. Conditional causal claims with more hedges and less certainty drop sharply in summaries. Example claim changes are presented which move from associational in the abstracts to causal in the summaries for each prompt except 'careful', where the reverse is displayed.

What might explain the widespread use of causal language observed in this study? Although we do not test mechanisms, we speculate that several interdependent factors probably contribute. First, universities' hiring and promotion practices heavily prioritize the number and status of candidates' publications^{53,54}. Second, journals, especially the highly selective journals that are most prized by hiring and promotion committees, often explicitly select for 'novel' and 'important' findings—criteria that encourage rhetorical overclaiming^{55–58}. Our finding that higher-impact journals contain more causal language is consistent with this interpretation. Third, funding agencies' increasing emphasis on 'impact'^{59,60} may also incentivize causal framing as a way to signal relevance. Fourth, questionable research practices remain widespread in the social sciences^{61,62}, potentially normalizing overstatement and reducing individual incentives to resist. Finally, our results are consistent with prior findings that humans routinely infer causality from correlation^{45–47}, a tendency that may stem from a fundamental cognitive preference for causal, narrative explanations⁶³. Scholars are presumably not immune to these biases⁶⁴, and this inclination may intensify by professional

incentives or ideological commitments^{7,65}. As a result, when researchers hold strong theoretical priors, resisting causal interpretation can be extremely challenging, demanding cognitive effort and sustained, deliberate attention to avoid.

Combating these mutually reinforcing mechanisms poses a formidable challenge, though publishing reforms offer perhaps the most obvious potential remedies. For example, *The Journal of Quantitative Description: Digital Media*, which explicitly discourages causal language, exhibited substantially lower rates of causal claims than most outlets. Ironically, this very observation illustrates the challenge we aim to highlight with this study: this finding is correlational and may reflect other mechanisms entirely (for example, self-selection among authors focused on more careful claims or editors already predisposed to methodological precision) yet it invites causal interpretation anyway. Nonetheless, we believe that clearer guidance from journals regarding permissible language could help realign scientific communication with methodological reality^{7,66}, provided such standards were consistently enforced rather than merely stated. Our results also suggest that LLMs, while potentially a part of the problem, can also be part of the

Table 2 | Practices of narrative license within social science articles

Practice	Definition	Examples
Selective reporting	Authors emphasize findings that support their narrative while minimizing or omitting conflicting evidence	Citing only literature that supports the hypothesis Highlighting aligned results, downplaying others Entirely omitting complicating results
Unreported deviation	Changes to methods or analysis plans are made without disclosure	Switching hypotheses after seeing results Polishing or simplifying methods without transparency Deviating from a preregistered plan without explanation
Rhetorical flourishes	Language is used to enhance appeal or obscure limitations	‘Hype’ or ‘boosting’ language Minimizing limitations through vague acknowledgements Adding excitement or certainty through editorializing
Overgeneralizing	Claims extend beyond what the data directly support	Applying findings to broader populations than studied Generalizing results to different settings or timeframes Treating specific measures as if they reflect broad concepts
Overclaiming	Authors draw stronger conclusions than the evidence justifies	Interpreting statistical significance as practical impact Describing small effects as robust or definitive Inferring causal relationships from correlational designs*

*In this analysis, we focus on causal language within correlational designs, but this is just one form of rhetorical overreach or narrative license, which may appear in academic publications.

solution. For example, by flagging instances of causal overreaching, LLM-powered tools can assist authors, reviewers, editors and readers alike to more clearly communicate and more accurately interpret scientific findings.

Before concluding, we emphasize that our study has several important methodological limitations, which also serve as motivation for future research directions.

First, our focus on inappropriate causal language is just one manifestation of verbal misrepresentation. As presented in Table 2, researchers have many rhetorical tools that can subtly distort findings—a broader phenomenon that we call ‘narrative license’, which captures the tendency to shape results into compelling stories at the expense of accuracy. Narrative license closely resembles prior work on ‘spin’⁴, but foregrounds narrative as the primary underlying mechanism. Much of the spin literature has focused on presenting null results as positive^{67,68}, but other forms have been examined^{35,51,69,70} and found to alter reader judgements^{71,72}. Future research might leverage LLMs to systematically detect other forms of narrative license at scale. Hedging and certainty within claims represent a particularly promising target: both have been extensively studied^{73–75} and validated detection methods already exist for scientific claims⁷⁶. Preliminary analyses suggest that certainty metrics are only moderately correlated with causal overclaiming (‘Causal claims and uncertainty’ section in the Supplementary Information), indicating that these rhetorical elements operate somewhat independently, reinforcing the value of examining different types of narrative license.

Second, our focus on cross-sectional studies, although helpful for creating clear criteria for identifying instances of causal overreach, does not imply that other designs are less susceptible to narrative license. For example, carefully executed cross-sectional evidence can, in some settings, be more informative for theory testing than, say, laboratory experiments with limited external validity. Indeed, randomized controlled trials carry a host of methodological limitations

that constrain the inferential scope of their findings^{6,77}. Our aim is not to criticize any single design or to single out any one manifestation of overclaiming, but rather to call attention to the broader phenomenon of narrative license and to introduce methods for identifying it. Future work should therefore examine other forms of narrative license within other research designs. If systemic pressures such as funders’ emphasis on ‘impact’ or publication incentives are driving these trends, we would expect parallel increases in other modes of overstatement catalogued in Table 2.

Third, in this study, we treated all causal claims about novel empirical results from cross-sectional studies as overreach. However, some such correlations do reflect genuine causal relationships, and ignoring these causal signals can also be harmful, as illustrated by Ronald Fisher’s scepticism about the smoking-cancer link, which was rooted in the observational nature of the evidence^{78,79}. Furthermore, overstated claims can occasionally accelerate progress, by provoking methodological reform or rapid follow-up work, as in the Cowles Foundation’s econometric contributions or the LK-99, ESP and cold-fusion episodes^{80–82}. We also note that humans are, at their core, storytelling creatures^{63,83}, who understand and retain information more effectively when it is embedded in narrative form. Eliminating narrative altogether from science communication is therefore not only probably impossible but also probably counterproductive. Although we continue to argue that caution when making causal claims is warranted in science communication, future work should consider how to balance the harms caused by careless rhetoric against the harms caused by excessive scepticism.

Fourth, our human-subjects experiment targeted college-educated participants but did not, to our knowledge, include practicing social scientists. Prior work on statistical misperceptions⁸⁴ shows that even expert judgements vary with how data are presented, but experts might respond differently in this specific case. Relatedly, our experiment focused on titles and abstracts, which omit context that careful readers might glean from full texts. Although our second LLM experiment found no evidence of a significant difference between abstract-only and full-text inputs, future human-subject studies should examine whether full-text exposure affects causal interpretations. Moreover, our experiments with language models reflect only a few current models under four prompt configurations. Other models, prompts or parameters may yield different outcomes, and we encourage continued evaluation of rhetorical overclaiming as new systems are deployed. Indeed, GPT-5.4 no longer showed the pattern observed in GPT-4.1; further testing will be necessary to determine whether this trend will persist in future models. Future studies should also examine whether LLMs similarly modify claims in more naturalistic usage contexts, such as multi-turn interactions or ‘deep research’ prompts on targeted topics, where models must select, integrate and summarize evidence across multiple sources.

Fifth, our study deliberately focused on academic publications as that is where practicing scientists have the most control and, hence, the most accountability, over messaging. However, science communication unfolds across multiple interacting stages⁸⁵, for instance, spin in journal articles can shape press releases⁸⁶, which in turn affect the accuracy of media reports⁸⁷. With recent advances in LLMs, there is substantial opportunity to scale fidelity detection from prior work⁸⁸. Future research could deploy LLMs across the full science communication pipeline to flag misaligned claims and better align interpretations with underlying evidence.

In conclusion, constructing coherent narratives around empirical findings can make it difficult to avoid causal overreach and related forms of overclaiming. Resisting these pressures requires sustained attention to what Feynman described as the first principle of scientific integrity: not to fool ourselves, despite being the easiest ones to fool⁸⁹. This intellectual discipline is an essential task required to ensure the integrity of science and the gradual accumulation of knowledge.

Methods

We present our methodology in two parts: first, the procedures for the descriptive analysis of academic articles; second, the methods used in the preregistered human-subjects experiment and experiments with LLMs.

Ethical considerations

For the preregistered, human-subjects experiment, the study protocol was reviewed and approved by the University of Pennsylvania's Institutional Review Board (IRB #858559). Participants provided informed consent electronically and could withdraw at any time. Data were collected anonymously and stored in compliance with IRB guidelines. Participants were compensated US\$2.25, which slightly exceeded Prolific's fair compensation guidelines for the estimated study duration. The experiments with LLMs and the descriptive analysis were not preregistered.

Methods for descriptive analysis

Open Science statement. This observational study was not preregistered. We share all analysis code, prompts and derived article-level datasets (methodology and causal-language labels, metadata and identifiers) on OSF (<https://osf.io/zpua6/>). The underlying article texts are licensed through ProQuest TDM Studio and cannot be redistributed; however, our repository includes the exact search strategy and processing pipeline so that researchers with institutional access can reconstruct the corpus and replicate our analyses.

Data and sample. This study draws primarily on data obtained through ProQuest's TDM Studio, a text and data mining platform that provides access to the full XML content of articles across several ProQuest databases. We supplemented this dataset with articles from elite journals not available within the five databases analysed (described below). As our main analyses focus on the TDM Studio articles, we describe that dataset first.

TDM Studio operates within a secure, browser-based workbench (often referred to as a 'walled garden'), which enables large-scale text mining of academic and journalistic content. The XML files include the title and abstract for all academic articles and, occasionally, full-text content as well. Within TDM Studio, we limited our search to five ProQuest databases that collectively represent a range of disciplines within the social sciences:

- ABI/INFORM Global provides comprehensive coverage of business, management and economics research, including scholarly journals, trade publications and market reports.
- Worldwide Political Science Abstracts indexes scholarly literature in political science, international relations, public policy and related disciplines, offering insights into global governance and political theory.
- International Bibliography of the Social Sciences is a multidisciplinary database covering economics, sociology, anthropology and political science, with an emphasis on international and interdisciplinary research.
- PAIS Index focuses on public policy, international relations and social issues, indexing government documents, think tank reports and academic literature on political and social affairs.
- Sociological Abstracts provides access to sociological research, including studies on social behaviour, culture, demography and social institutions.

To identify potential cross-sectional studies, we queried for relevant keywords ('cross-sectional' OR 'crosssectional' OR 'survey' OR 'observational study' OR 'prevalence study' OR 'questionnaire') within the abstracts of all articles in these databases. This returned 672,944 total articles. ProQuest labels whether articles came from scholarly journals. Roughly half of the articles contained that classification (48.7%, $N = 327,598$). Our primary analysis is conducted on these scholarly articles that passed peer review; however, we include additional analysis on the full set of articles, which includes working papers and

preprints, dissertations and theses, trade magazines and conference submissions ('Results for non-academic journals' section in the Supplementary Information). Although each article had a unique identifier within these databases, a small portion (1.9%, $N = 6,276$) contained duplicate titles. These duplicates were dropped from our analysis, leaving a final sample size of 321,322 unique article abstracts. Figure 1 shows a visualization of how articles were collected and analysed.

Automatic detection of study methodology and causal language.

To identify cross-sectional studies and determine whether they used causal language, we developed a computational approach designed to operate within the constraints of the TDM Studio platform. A key limitation of this 'walled-garden' environment is that all processing must be conducted locally; it does not allow for easy sharing of content among coders or large-scale use of language models within its infrastructure (note that after our analyses were conducted, TDM Studio opened access to OpenAI's API.). However, TDM Studio does support the export of small batches of metadata, including titles and abstracts, which enabled limited external processing. Because all articles contained this metadata but only a subset contained full texts, we needed to build classifiers that relied solely on titles and abstracts.

To construct these classifiers, we followed the steps outlined in Fig. 1. First, we developed coding procedures for humans and corresponding prompts for LLMs to identify two features: (1) whether the study used a cross-sectional design and (2) whether it used causal language in describing its results. Human coders then rated a sample of 200 titles and abstracts to validate these classifications, alongside ratings from GPT-4o. We used this validation set to assess the performance of the language model. After validation, we applied the LLM to a larger dataset ($N = 30,000$) and used its outputs to train a local BERT-based classifier that could approximate the LLM's judgements. This classifier was then validated on a held-out sample ($N = 10,000$) and then deployed within TDM Studio to classify the full dataset. This approach allowed us to harness the contextual understanding of LLMs while operating within the platform's technical constraints. Both the language model and the local classifier achieved strong accuracy for both classification tasks, as described in detail below.

Identifying cross-sectional articles. Cross-sectional studies are observational research designs that examine empirical data collected from different entities. A key feature is that they do not track repeated measurements and do not involve any (quasi-)experimental intervention. To identify such studies, human coders independently reviewed the methodological design of 50 articles, consulting full-text content when abstracts were ambiguous ('Instructions for human coding of methodology' section in the Supplementary Information). Ultimately, each article was assigned a binary label: cross-sectional or non-cross-sectional. The coders achieved acceptable agreement (Krippendorff's $\alpha = 0.84$) and resolved discrepancies through discussion to reach a consensus on final labels. They then coded an additional 150 articles to create the validation set of 200 randomly selected articles.

Using our coding instructions as a foundation, we developed a corresponding prompt for GPT-4o to classify study methodology (Extended Data Box 1). On our held-out validation set of 200 articles, the model showed strong alignment with human judgements (F1, 0.87; precision, 0.84; recall, 0.91; accuracy: 0.85). We applied this prompt to the 30,000 titles and abstracts exported from TDM Studio and used these labels to train a BERT-based classifier ('Training BERT-based models' section in the Supplementary Information). The classifier demonstrated strong predictive performance on a held-out validation set of 10,000 scholarly journal articles (F1, 0.92; precision, 0.91; recall, 0.93; accuracy, 0.91), indicating that it closely approximates GPT-4o's judgements and is suitable for large-scale automated coding. The trained model was then uploaded to TDM Studio and applied to the full dataset. Of the 321,322 unique, scholarly articles retrieved through our

query, the classifier identified 194,631 as using cross-sectional designs. All subsequent analyses were conducted on this subset.

Identifying causal language in titles and abstracts. To identify causal language in the abstracts of the 194,631 cross-sectional studies, we followed the same procedure used to classify study design. We began by randomly sampling 50 articles, which were independently assessed by three expert coders to determine whether the title or abstract contained causal language about the article's novel empirical findings, excluding claims regarding context or background literature (see 'Instructions for human coding of causal language' section in the Supplementary Information and Supplementary Table 1 for examples). Causal language was defined broadly to include direct, implied and conditional forms. For direct causal language, the authors explicitly claim that their results show one variable affects or causes another in their title or abstract (for example, "We show that social media causes an increase in depressive symptoms among adolescents."). For implied statements, the abstract suggests that their study investigates a causal relationship (for example, "We explored the role of social media use in shaping depression outcomes.") or proposes an intervention that entails causality without listing their assumptions or hedging their implication (for example, "Based on our findings, individuals should limit their social media consumption to improve their mental health outcomes."). Conditional statements are hedged or qualified causal claims (for example, "Social media may contribute to depressive symptoms in teens."). Although conditional statements are technically justified, we included them because we believe that they may similarly lead to excessive causal inference compared with more careful phrasing. We note, however, that conditional claims are relatively rare: of the articles classified as causal by human coders, only 3% contained only conditional claims, whereas the remaining 97% contained either direct (54%) or implied (43%) claims (where articles contained more than one type of causal claims we classified them by the strongest form present). Note that correlational claims framed as aligned with causal hypotheses were not targeted by our classifier and were therefore generally marked as associational. Given the low prevalence of conditional claims and the modest agreement of human coders when using this four-level coding scheme (Krippendorff's $\alpha = 0.67$), our primary analysis relies on the simpler binary outcome: whether any form of causal language was present. For this binary classification, human coders demonstrated strong agreement (Krippendorff's $\alpha = 0.86$). Coders then classified 150 more articles to create the validation set of 200 randomly selected titles and abstracts, though ten articles were removed because they were in a language other than English.

Using our coding instructions, we developed a prompt for GPT-4o to classify the presence of causal language in an article's title and abstract (Extended Data Box 2). GPT-4o showed strong alignment with the human-coded labels on our held-out validation set (F1, 0.81; precision, 0.91; recall, 0.74; accuracy, 0.78). Notably, the model's higher precision than recall suggests it tends to miss some causal claims, leading to a slight underestimation of their prevalence compared with human judgements. We then applied the GPT-4o prompt to the set of 30,000 titles and abstracts exported from TDM Studio and used the resulting labels to train a BERT-based classifier. This model demonstrated strong predictive performance on the held-out sample of 6,027 cross-sectional articles drawn from the 10,000-article validation set (F1, 0.83; precision, 0.84; recall, 0.82; accuracy, 0.84), suggesting it effectively approximates GPT's judgements and is suitable for large-scale automated coding. The trained model was subsequently uploaded and deployed within the TDM Studio environment, allowing us to apply it to the full dataset of cross-sectional articles on the platform. Table 1 presents examples of causal language found in randomly sampled titles and abstracts from our full dataset ('Examples of causal and descriptive framing in abstracts' section in the Supplementary Information).

Robustness checks. To evaluate the sensitivity of our analysis to measurement strategy, we conducted two alternative approaches to identifying causal language, including keyword searches and a causal language model developed in prior research³⁵ and found even greater prevalence rates ('Alternative approaches to assessing causal language in titles and abstracts' section in the Supplementary Information).

Full text analysis. A subset of our cross-sectional articles from scholarly journals ($N = 69,580$, 35.7%) included full text in the XML. Although these articles are not randomly distributed, we aimed to determine whether their full texts were more or less likely to contain causal language compared with titles and abstracts. We required a solution that could be implemented in TDM Studio.

To achieve this, we first searched the full-text documents for a set of causal keywords ('Full text analysis' section in the Supplementary Information). When a keyword was identified, we extracted a snippet of text spanning 100 characters on either side of the keyword to capture the relevant context describing the outcome. We opted for snippets rather than complete sentences because the XML data are highly variable and often too messy for sentence classifiers to work properly. Moreover, additional context is sometimes necessary to establish whether the causal language is relevant to new results or instead comes from previous work. Next, we used regex to detect citation patterns within these snippets. Any snippet containing a citation was excluded from further analysis. Finally, we applied our BERT-based causal language identifier to these snippets.

Full text claim-level analysis. The primary abstract-level analysis was limited in that it relied on a single classification per abstract, even though abstracts often contain multiple claims, and full texts contain many more. This approach also conflated distinct forms of causal language (collapsing conditional with implied and direct causal claims). Although conditional causal claims represented only a small portion of the dataset, we acknowledge that such claims are more justifiable than direct or implied ones and, ideally, should be analysed separately. It is therefore important to assess how frequently full-text claims use conditional (hedged) versus stronger statements that explicitly assert a causal relationship.

To address these issues, we conducted an additional analysis on a curated sample ($N = 100$) of articles representing different levels of causal strength, breaking them down into individual claims to allow multiple labels per article. To construct this sample, we randomly selected abstracts from the 30,000 originally exported from TDM Studio that had been classified by GPT into four categories (descriptive > conditional > implied > direct; 'Results by causal type' section in the Supplementary Information). Two authors reviewed these abstracts to verify the accuracy of GPT's classification, with the final sample consisting of 25 articles for each category where GPT and the coders showed complete agreement (total $N = 100$).

To analyse individual claims, we examined sentences as a smaller unit of analysis. Abstracts contain sentences that serve multiple functions, including providing background, stating the research objective, describing methods, presenting results and articulating implications. We split each abstract into sentences using the sentence tokenizer in the python spaCy package⁹⁰, classified all sentences into one of those categories and restricted our analysis to those classified as about the results. This approach is conservative: it omits all implied claims made in the research objective as well as those that appear in the implications. Thus, any causal claims remaining in this analysis should explicitly concern the novel findings reported by the authors. Note that there were three abstracts that did not contain any sentences whose primary label was results; for these, we selected sentences that combined results with implications or results with background.

Using the Scientific Information Change Package⁹¹, which was developed to measure the degree of similarity in the description of scientific findings, we compared claims across abstracts and main texts. Following

the procedure described by Wright et al.⁹¹, we classified pairs of claims with an information-matching score greater than 3.0 as representing the same finding. We retained only the first occurrence of each paired claim from the abstract, removing duplicates. This process yielded a total of 182 claims across 100 abstracts, with the number of claims per abstract varying (mean of 1.82, s.d. of 1.1; minimum 1, maximum 6). We then applied our fine-tuned BERT-based classifier to determine whether each claim was causal or noncausal. In addition, we assessed the presence of hedging terms ('may', 'would', 'can', 'might', 'could', 'appear to', 'seem', 'likely', 'potential', 'possible', 'indicative', 'probable', 'suggest', 'support', 'model', 'relationship', derived from ref. 92). Claims classified as causal without a hedge were classified as unhedged causal, causal claims with a hedge were classified as conditional causal and noncausal statements were labelled associational/descriptive. Because we focused only on the results sentences, we removed the implied causal category.

This analysis revealed that many claims within each abstract-level causal category were associational or descriptive, although causal claims were more frequent in abstracts previously classified as causal across types. Instances of unhedged causal claims appearing in descriptive, conditional or implied abstracts often arose when a sentence appeared associational in context but seemed causal when viewed in isolation. For example, the statement "Our results show that negative returns to health outcomes set in at around 50 work hours per week, and that the negative effects of working long hours manifest earlier for women than men." immediately precedes a sentence clarifying the associational nature of the finding. Although in context, the statement is arguably noncausal, when analysed in isolation, its phrasing ('set in', 'manifest') was classified as an unhedged causal claim by our classifier. This example illustrates a central tradeoff in the sentence-based, claim-level classification: although it allows for greater granularity, it can obscure contextual nuance. The breakdown of claim classifications by abstract-level category is presented in Supplementary Fig. 4.

Next, we examined paired claims within the full text. We first gathered the full texts for all articles in the above analysis and manually divided each article into four sections: introduction, methods, results and discussion. Not all journals adhere strictly to this structure—some intersperse discussion within the results, and others place descriptive results in the methods section—so we used our judgement to assign text to the section that best fit its function. For each section, we followed a procedure parallel to the abstract-level analysis. We tokenized the text into sentences, then used regex patterns to remove any sentence containing references to prior work, to focus on sentences describing the current study. We then matched all remaining claims to the unique abstract claims using the Scientific Information Change package, retaining all pairs with an information-matching score greater than 3.0. This generally yielded multiple matches for each abstract claim, with substantial variability (total $N = 3,134$; median of 11, mean of 18.1, s.d. of 18.4; minimum 1, maximum 100). These matches were unevenly distributed across sections: the introduction (mean of 6.6) and discussion (mean of 6.3) contained more claims per abstract claim than the results (mean of 4.2) or methods (mean of 1.0). For our main full-text analysis, we retained all unique matched claims. We applied our causal language classifier and hedge detector to all matched full-text claims, categorizing each as unhedged causal, conditional causal or associational/descriptive.

We note that this approach has some important limitations. First, by focusing only on explicit results claims, we overlook subtler ways in which authors imply causality in their framing of the research question or in their proposed implications, which may lead us to underestimate the prevalence of causal language relative to methods that incorporate these claim types. Second, because the claim-level analysis is conducted at the sentence level, it can sometimes miss contextual cues indicating that the results are associational or providing important limitations or caveats. These limitations are addressed with our abstract-level classification, indicating the importance of multiple approaches to analysis and providing convergent validity for the prevalence of overclaiming.

Refer to the 'Full text claim-level analysis: additional results' section in the Supplementary Information for additional details.

Supplemental data. We gathered information about every journal in our dataset using two complementary approaches. First, we used GPT-4o to classify the disciplinary focus of each journal based on the publication title ('GPT-4o prompt to label journal discipline' section in the Supplementary Information). Second, we matched journal titles to the SJR database, a widely used platform that provides detailed bibliometric data for academic journals. The SJR database covered a substantial portion of the journals in our dataset ($N = 127,598$ cross-sectional articles), allowing us to incorporate standardized impact metrics in our analysis.

In addition to the TDM Studio dataset, we gathered articles from several high-impact general science journals that were not included in the ProQuest databases. Specifically, we collected content from *PNAS*, *Nature*, *Nature Human Behaviour*, *Science* and *Science Advances* ('External data from elite venues' section in the Supplementary Information). For each journal, we filtered articles to ensure they focused on the social and behavioural sciences and included one of the keywords used to identify articles on TDM Studio (final $N = 454$). Note that our scraping of *PNAS* also includes articles published in *PNAS Nexus* following its launch in 2022. These sources helped ensure broader disciplinary coverage and representation of methodological variation within the behavioural sciences.

Given the prominence of these journals and their rigorous review processes, we conducted additional validation, both for them and for 15 other prestigious disciplinary venues included in TDM Studio, to ensure accurate identification of cross-sectional designs and causal language ('Validating elite journals' section in the Supplementary Information). Because the sample from these venues was relatively small ($N = 3,713$ abstracts), we used GPT-4o for classification. Then, we drew a stratified random sample of up to four cross-sectional articles per venue and had two authors manually review the full text, confirming that the majority (68%) met strictly cross-sectional criteria. These coders also labelled the presence of causal language in abstracts (blind to GPT's labels), and their judgements showed strong agreement with our automatic classifier (F1, 0.82; precision, 0.91; recall, 0.75; accuracy, 0.82), with notably higher precision than recall, indicating that GPT's labelling underestimates the rate of causal language relative to human judgements. Finally, we incorporated articles ($N = 101$) from a specialized journal with editorial policies that explicitly target limiting causal language in descriptive scientific work (*Journal of Quantitative Description: Digital Media*).

Methods for experiments

The human-subjects experiment examined the extent to which readers inferred causality from cross-sectional research abstracts that contained overreaching causal language, as identified by our classifiers in the descriptive analysis. In addition, we tested three interventions: (1) a rewrite condition in which the abstracts were revised to retain the same content but use only associational language, (2) a simple methodological label clarifying that the study was cross-sectional and does not establish causality and (3) an 'AI' feedback condition that combined this label with brief commentary and in-text highlights that drew attention to key methodological and interpretive features of the abstract. We were interested in the following research question and hypotheses:

- (1) To what extent do participants judge that a study provides causal evidence after reading a cross-sectional abstract that uses causal language to describe its results?
- (2) Each of the three interventions—a rewritten abstract using purely associational language, a methodological note clarifying study design and limitations and 'AI' feedback highlighting interpretive language alongside a similar note—will reduce the amount that participants report a study provides causal evidence.

Open Science statement. All procedures, materials, hypotheses and analytic plans for this experiment were preregistered on AsPredicted (registration #238174; <https://aspredicted.org/rhz3-jtg5.pdf>) on the morning of 14 July 2025, before data collection, which began later that day. We followed this plan exactly, with no deviations from the protocol except for minor overrecruitment by the platform. Deidentified data, analysis scripts and stimuli files are deposited on the Open Science Framework for full reproduction of our results (<https://osf.io/zpua6/>). The experiments with LLMs were not preregistered.

Design. We recruited 1,100 adult participants through Prolific using the Wharton Behavioral Lab. Eligibility was restricted to individuals who hold at least a bachelor's degree and reside in the USA. A priori power analysis indicated that this sample size achieved over 80% power of detecting a modest effect size conservatively estimated based on pilot data results ('Power analysis' section in the Supplementary Information). The platform overrecruited an additional five participants, resulting in a final $N = 1,105$. We did not collect demographic information about participants' age and sex.

The experiment used a between subjects design with four main conditions. The factors were: (1) original: the title and abstract containing causal phrasing; (2) rewrite: rewritten title and abstract using strictly associational phrasing; (3) methodological note: a brief note highlighting the cross-sectional methodology with the original title and abstract; (4) 'AI' feedback: the methodological label above plus brief 'AI' feedback and inline highlights that flag key areas which may impact reader inferences. Although participants were informed that this was AI generated, we manually created this label and the highlights to ensure consistency across articles. Participants were randomly assigned to one of the four cells. See Extended Data Fig. 5 for an illustration of the experiment design. We also included an exploratory 'No Info' condition for certain outcomes that did not require reading the study abstract. We randomized participants to conditions with 216 in the original, 240 in the rewrite, 229 in the note, 202 in the AI and 218 in the 'No Info' condition.

We created a pool of 28 cross sectional research abstracts drawn from our manual validation of articles in top tier journals ('Validating elite journals' section in the Supplementary Information). Articles were selected through stratified random sampling to ensure disciplinary diversity, and all original articles with both a cross-sectional methodology and any causal claim in their title or abstract were utilized as stimuli. For each article we prepared two matched versions, the original title and abstract and a rewritten version that used purely associational language, while keeping the content as similar as possible otherwise ('Experiment stimuli' section in the Supplementary Information).

The primary dependent variable was participants' averaged responses to two questions about whether the study provided causal evidence: the first tailored to the abstract they read, and the second stated more abstractly, "Based solely on the evidence presented in this paper (not your prior beliefs or intuition), do you agree that this paper shows that {changes in independent variable} directly causes {changes in the dependent variable}? (1 Strongly disagree - 7 Strongly agree)"; "Do you agree that the study design used in this paper can establish a causal relationship between the variables of interest? (1 Strongly disagree - 7 Strongly agree)". The specific independent variable and dependent variable terms were populated from the abstract assigned to each participant, ensuring the language was consistent with how the variables were described in the text.

In addition to our primary outcome, we collected two secondary outcome variables: (1) free-response item: "In a few sentences (at least 25 words) of your own words, summarize what this study shows about {independent variable} and {dependent variable}". Responses to this question were processed using a slightly updated causal language classification prompt with minor updates for these summaries, yielding a binary indicator of whether the participant used causal phrasing ('First

experiment with GPT-4.1' prompt in the Supplementary Information). (2) Subtle and implied causal inference: averaged responses to two items, one about a hypothetical behaviour/policy intervention and another about their personal beliefs: "Given this information and using your intuition, do you agree that if {group changes X} we can expect {Y to change} as a result? (1 Strongly disagree - 7 Strongly agree)"; "Based on your intuition and the information provided, do you agree with the claim that {changes in independent variable} directly cause {changes in dependent variable}? (1 Strongly disagree - 7 Strongly agree)" ('Experiment stimuli' section in the Supplementary Information).

Procedure. After consenting, participants read brief background information about one randomly selected study, which provided definitions and other relevant details for comprehension. They were then provided with the associated title and abstract. Formatting (font, size) was identical across conditions, and all abstracts were PNGs to prevent copy and pasting. They then answered the outcome questions, always completing the summary first to ensure greater engagement, and then the other items were presented following a Latin square to account for order effects. They next provided demographic information about their level of (statistical) education. Finally, they had the option to leave comments before exiting the survey. Participants in the note and AI condition also answered a manipulation check question about the message, and participants in the exploratory 'No Info' control condition also answered the subtle and implied causality questions immediately after the background but before reading the study abstract. This design allowed us to assess changes in participants' implied causal beliefs using measures collected both before and after exposure to the abstract.

Participants who failed the attention check or submitted responses in under one third the median pilot time were flagged. Consistent with the preregistration, no participants with complete data were excluded from the main analysis; flagged cases were retained and examined for robustness in sensitivity analyses ('Sensitivity analysis without flagged participants' section in the Supplementary Information). Median completion time was 8 min and 32 s; 11 participants were below the cutoff time, and an additional 25 participants failed the attention check, resulting in a total of 36 participant responses that were flagged (3.3%).

Statistical analysis. Analyses were conducted in R. Our primary model was a multilevel linear regression with random intercepts for article (equation (1)). We were principally interested in main effects, though we report additional moderation analysis in the supplement ('Exploratory analysis for the experiment' section in the Supplementary Information). Our primary dependent variable was the average response to the two questions about whether the study provided causal evidence, and we ran a parallel model with the subtle and implied causality questions as well as a similar logistic regression with the binary classification of whether the summaries contained causal claims

$$Y_{ij} = \beta_0 + \beta_1 W_{ij} + \beta_2 L_{1ij} + \beta_3 L_{2ij} + u_j + \epsilon_{ij}, \quad (1)$$

where Y_{ij} is study provides causal evidence for participant i reading article j ; W_{ij} is the rewrite condition (where 0 is original, 1 is rewritten); L_{1ij} is the methodological note condition (where 0 is no note, 1 is note present); L_{2ij} is the AI condition (where 0 is not AI, 1 is AI condition); u_j is the random intercept for article j ; and ϵ_{ij} is the residual error.

Model assumptions were verified via residual plots, and the main results were found to be similar across specifications ('Residual diagnostics' section in the Supplementary Information). All preregistered hypotheses were evaluated with one tailed tests at $\alpha = 0.05$ with a Holm–Bonferroni correction for multiple comparisons within each model family. In R, we used the lme4 package for mixed-effects modelling⁹³. In Python, we used statsmodels for statistical modelling and matplotlib for visualization^{94,95}. Data collection and analysis were not performed blind to the conditions of the experiments.

Exploratory analysis. We were also interested in how both features of the stimuli and characteristics of the participants moderated causal inference. Specifically, we explored the plausibility of the causal relationship from each abstract (as rated by language models), the strength of causal language used in the original version of the abstract (as rated by statistical experts) and the complexity of the abstract's language, including both Flesch–Kincaid reading level and subjective expert judgements. We also considered participants' statistical education, based on a self-report question. We examined how these variables related to our three outcome measures using multilevel models with interaction terms ('Exploratory analysis for the experiment' section in the Supplementary Information).

First experiment with GPT-4.1. Given the growing role of AI in both consuming and generating social science content, we conducted a parallel experiment on GPT-4.1, in which this LLM was presented with the same set of abstracts and questions as human participants. We gathered ten responses from the AI agent for every article and condition and report these responses alongside those of human readers for comparison ('First experiment with GPT-4.1' section in the Supplementary Information). When querying the model, we used temperature of 1.0 and the default presence penalty (0). For multiple-choice items, max_tokens = 5; for summaries, max_tokens = 100. All other parameters were left at July 2025 batch-API defaults. Importantly, we do not assess AI judgements based on how closely they resemble human responses; rather, we examine them in their own right, given the increasing role of language models as mediators and communicators of scientific information.

Second experiment with LLMs. Our second experiment with LLMs explored whether the probability of a direct causal claim appearing in model summaries varied by the abstract-level causal classification, the summarization prompt and whether the model was provided with the full text rather than only the abstract.

For our sample, we used the stratified sample of 100 articles from the 'Full Text Claim Level Analysis', which were evenly divided by abstract-level causal classification (direct, implied, conditional, associational/descriptive). This design allowed us to test whether models added stronger causal language for articles whose abstracts were conditional and whether they introduced causal claims when abstracts were purely associational from the start. We created four prompts representing diverse real-world uses of language models for interpreting academic texts: (1) a basic summarization prompt, (2) a careful prompt encouraging caution and scepticism, (3) a simplified prompt requesting a summary at an eighth-grade reading level and (4) a practical prompt emphasizing real-world significance ('Second experiment with LLMs—prompts' section in the Supplementary Information).

For each article and prompt, we submitted two requests to each model: one containing only the title and abstract and another containing the full text, to test whether access to additional context affected causal framing in summaries.

We began with only GPT-4.1. Each request used a temperature of 1, a maximum of 150 tokens and generated five outputs, with all other parameters set to the default GPT-4.1 Batch API settings as of 4 November 2025. Because new leading models were deployed throughout the peer review process, we replicated these results with subsequent models to test generalizability: specifically, OpenAI's GPT-5.1 and GPT-5.4, DeepSeek Chat-V3.2 and Anthropic's Claude Sonnet 4.6. For these models, we returned a single summary per query via the chat API and collected all responses from 25–28 March 2026.

For our outcome measure, we developed a new classifier tailored to these simplified summaries. As with our full text analysis, this classifier did not target the implied category, and we also relabelled the direct category as 'unhedged causal', resulting in three categories (unhedged causal > conditional causal > associational/descriptive). Each output summary received a single label corresponding to the

strongest causal claim type present, and the model classifier showed moderate alignment with the consensus human label on a held-out validation sample of summaries (macro $F_1 = 0.75$, precision of 0.75, recall of 0.79, accuracy of 0.75; $N = 240$; 'Second experiment with LLMs: causal classifier' section in the Supplementary Information). Notably, in cases where models and the consensus expert judgement mismatched, human coders were substantially more likely to label summaries as containing an unhedged causal claim than the model (Z test of proportions = 3.3, $P = 0.0008$). There were 44 cases where the humans labelled a summary as 'causal' when the model's label was either conditional or associational and only eight with the reverse. For GPT-4.1 responses, we aggregated across the five outputs and assigned the majority label to each article–prompt–content type; in cases of ties, we randomly selected one label from the tied categories. For all other models where we collected only one response, we assigned that label only.

For an analytical approach, we first assessed whether summaries generated from full texts versus abstracts differed systematically in their causal classifications. The summary coding scheme included three categories of causal language (that is, descriptive, conditional, unhedged). To preserve this structure, we report overall agreement, Cohen's κ and P values from Stuart–Maxwell tests to evaluate whether the marginal distributions of causal classifications differed across content types. We report both raw and false discovery rate-adjusted P values, analysing every prompt–model configuration separately. These statistics are presented in Supplementary Table 7. Agreement statistics should be interpreted alongside these distributional comparisons, as κ can be sensitive to differences in category prevalence.

For the main analysis, we collapsed the coding scheme into a binary distinction of whether the summary contained an unhedged (that is, direct) causal language or not. This focuses the analysis on the most substantively consequential form of overreach. We then assessed whether model-generated summaries were more likely to contain such overstatements relative to the original abstracts. Because summaries generated from full texts and abstracts did not differ meaningfully, we restrict this section to abstract-based summaries to avoid double-counting.

We used complementary analytical approaches. First, we present a complete set of 2×2 contingency tables comparing causal classifications in abstracts and summaries for every model/prompt pair (Supplementary Table 8). For each configuration, we report overall agreement, differences in the proportion of summaries classified as containing unhedged causal language and results from McNemar's test, including both raw and false discovery rate-adjusted P values (Extended Data Table 3). These analyses provide a condition-specific assessment of whether certain models or prompts tend to inflate or attenuate causal overstatement relative to abstracts. Given the nested structure of the data, with multiple observations derived from the same article, assessing the statistical significance of these differences across models and prompts is complex. Accordingly, these results should be interpreted in conjunction with the multilevel models reported below, which explicitly account for this clustering.

To assess systematic differences across models and prompts and to examine how causal language in abstracts relates to causal language in summaries, we estimated a multilevel logistic regression model using the lme4 package in R. In this model, the presence of unhedged causal language in the summary was predicted by the model, the prompt, their interaction and the abstract's causal classification (descriptive, conditional, implied, direct). All predictors were treated as factors. We included a random intercept for article to account for repeated observations. Model estimates are reported in Supplementary Table 9. Importantly, although this model enables clear comparisons across models, prompts and abstract causal categories, it does not permit direct comparisons of causal language rates relative to the original abstracts, which is why we also reported the 2×2 contingency analyses described above.

We summarize these results at a high level in the main text, providing point estimates with confidence bands only in the Supplementary Information due to space constraints.

We also conducted several robustness checks. To examine whether our results depended on our method for identifying causal language, we conducted sentence-level analyses on the responses produced by GPT-4.1, using our causal language classifier from the descriptive analysis, an independent BERT-based classifier trained on article conclusions³⁵ and a hedge detector, paralleling the procedures used in our full-text analysis ('Second LLM experiment: GPT-4.1 robustness checks' section in the Supplementary Information), finding similar results. We also tested whether abstract claim mix (descriptive versus unhedged causal) predicts causal language in LLM summaries and found that more causal language in the article yields more causal claims in summaries ('Second LLM experiment: GPT-4.1 claim level analysis' section in the Supplementary Information). Also refer to the 'Second experiment with LLMs: additional tables' in the Supplementary Information for details into the findings.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All article-level (including citation information) and experimental results datasets required to reproduce the findings of this study are available via the Open Science Framework (OSF) at <https://osf.io/zpua6/>. The underlying article texts and metadata were obtained through ProQuest TDM Studio and are subject to licensing restrictions. These data are therefore not redistributed by the authors but may be obtained directly via ProQuest's TDM Studio platform at <https://tdmstudio.proquest.com/home>, subject to institutional access and licensing agreements. Source data are provided with this paper.

Code availability

All analysis code required to reproduce the findings of this study is also available via the same OSF repository at <https://osf.io/zpua6/>.

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Author contributions

C.I., T.D., N.F., and D.J.W. conceived the study. C.I. and G.J. collected the data and conducted the analyses. C.I., T.D., N.F., G.J., and D.J.W. designed the experiment, which was conducted and analysed by C.I. C.I. and D.J.W. wrote the manuscript, and all authors reviewed and approved the final version.

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Competing interests

The authors declare no competing interests.

Additional information

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Extended Data Table 1 | Effects of Conditions on Human-Subjects Experiment Responses

Outcome	Condition	Statistic	<i>b</i> or OR	95% CI	β or MPP	p
Study provides causal evidence (1-7)	Original abstract (intercept b_0)	--	5.1	[4.86, 5.27]	--	--
Study provides causal evidence (1-7)	Rewrite	$t(878.2) = -2.7$	-0.4	[-0.63, -0.10]	-0.25	0.003
Study provides causal evidence (1-7)	Methodological note	$t(881.5) = -4.0$	-0.5	[-0.81, -0.28]	-0.37	<0.001
Study provides causal evidence (1-7)	AI feedback	$t(877.5) = -5.8$	-0.8	[-1.08, -0.53]	-0.55	<0.001
Subtler causal-inference outcome	Original abstract (intercept b_0)	--	5.3	[5.07, 5.52]	--	--
Subtler causal-inference outcome	Rewrite	$t(865.8) = -1.3$	-0.1	[-0.37, 0.07]	-0.12	0.092
Subtler causal-inference outcome	Methodological note	$t(869.1) = -1.7$	-0.2	[-0.42, 0.03]	-0.15	0.089
Subtler causal-inference outcome	AI feedback	$t(865.8) = -2.0$	-0.2	[-0.46, 0.00]	-0.18	0.073
Subtler causal-inference outcome	Original vs No-info baseline	$t(1081.3) = 0.8$	0.09	[-0.13, 0.30]	0.07	0.471
Subtler causal-inference outcome	Rewrite vs No-info baseline	$t(1078.0) = -0.6$	-0.07	[-0.28, 0.14]	-0.06	0.471
Summaries: presence of causal language	Original abstract (intercept odds)	--	2.7	[1.76, 4.10]	0.73	--
Summaries: presence of causal language	Rewrite	$z = -4.6$	0.38	[0.25, 0.57]	0.5	<0.001
Summaries: presence of causal language	Methodological note	$z = -1.4$	0.73	[0.48, 1.13]	0.66	0.078
Summaries: presence of causal language	AI feedback	$z = -2.3$	0.6	[0.39, 0.93]	0.62	0.022

Notes: Effects for rating outcomes are unstandardized coefficients (*b*) from linear mixed models. Effects for summary outcomes are odds ratios (OR) from logistic mixed models. The Statistic(df) column reports the relevant test statistic and associated degrees of freedom (e.g., $t(df)$ or z). The β column reports standardized effect sizes for linear mixed models and the Marginal Predicted Probabilities for the summary outcome. Intercepts (b_0) represent the original abstract condition. P-values are Holm--Bonferroni corrected within model families.

Extended Data Table 2 | Effects of Conditions on GPT-4.1 Responses for the first experiment

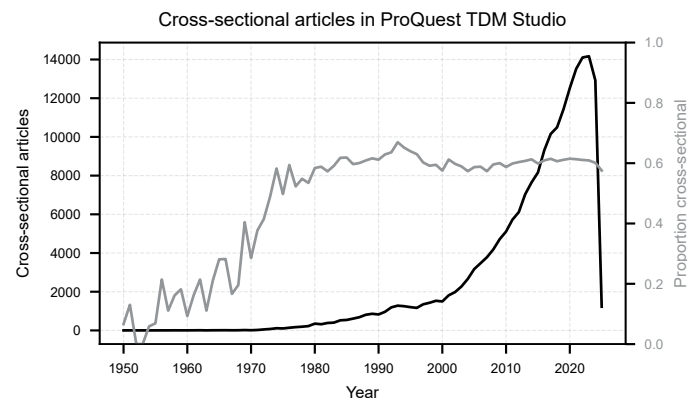
Outcome	Condition	Statistic	<i>b</i> or OR	95% CI	β or MPP	p
Study provides causal evidence (1-7)	Original abstract (intercept b_0)	--	3.1	[2.85, 3.38]	--	--
Study provides causal evidence (1-7)	Rewrite	$t(81) = -10.0$	-1.2	[-1.44, -0.97]	-1.23	<0.001
Study provides causal evidence (1-7)	Methodological note	$t(81) = -12.9$	-1.5	[-1.77, -1.31]	-1.58	<0.001
Study provides causal evidence (1-7)	AI feedback	$t(81) = -14.2$	-1.7	[-1.94, -1.47]	-1.75	<0.001
Subtler causal-inference outcome	Original abstract (intercept b_0)	--	5.8	[5.38, 6.19]	--	--
Subtler causal-inference outcome	Rewrite	$t(81) = -5.9$	-0.9	[-1.15, -0.58]	-0.6	<0.001
Subtler causal-inference outcome	Methodological note	$t(81) = -11.1$	-1.6	[-1.90, -1.33]	-1.13	<0.001
Subtler causal-inference outcome	AI feedback	$t(81) = -17.6$	-2.6	[-2.86, -2.29]	-1.79	<0.001
Subtler causal-inference outcome	Original vs No-info baseline	$t(108) = 6.5$	0.98	[0.69, 1.27]	0.7	<0.001
Subtler causal-inference outcome	Rewrite vs No-info baseline	$t(108) = 0.8$	0.11	[-0.18, 0.40]	0.08	0.447
Summaries: presence of causal language	Original abstract (intercept odds)	--	134.8	[45.4, 400.5]	0.99	--
Summaries: presence of causal language	Rewrite	$z = -5.4$	0.065	[0.02, 0.17]	0.9	<0.001
Summaries: presence of causal language	Methodological note	$z = -10.2$	0.005	[0.002, 0.015]	0.42	<0.001
Summaries: presence of causal language	AI feedback	$z = -11.0$	0.003	[0.001, 0.009]	0.3	<0.001

Notes: Effects for rating outcomes are unstandardized coefficients (*b*) from linear mixed models. Effects for summary outcomes are odds ratios (OR) from logistic mixed models. The Statistic(df) column reports the relevant test statistic and associated degrees of freedom (e.g., $t(df)$ or z). The β column reports standardized effect sizes for linear mixed models and the Marginal Predicted Probabilities for the summary outcome. Intercepts (b_0) represent the original abstract condition. P-values are uncorrected for this analysis.

Extended Data Table 3 | Comparison of Causal Language in Abstracts and LLM Summaries

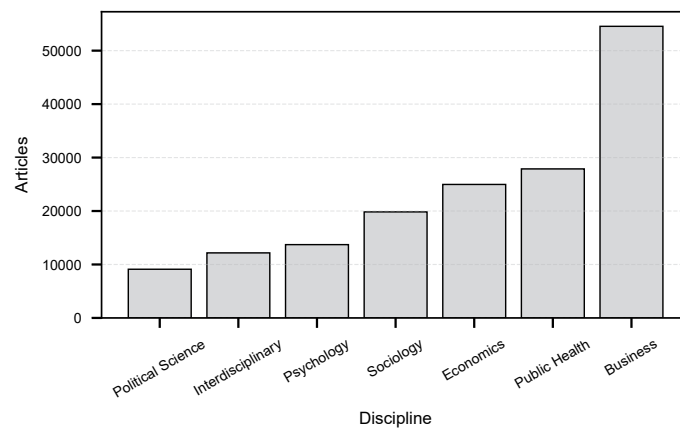
Model	Prompt	Agreement	Δ Proportion	95% Wald CI	McNemar p	McNemar p_{adj}
gpt-5.1	Basic	0.74	0.1	[0.00, 0.20]	0.0755	0.0809
gpt-5.1	Simplified	0.67	0.21	[0.11, 0.31]	0.0003	0.0006
gpt-5.1	Practical	0.71	0.17	[0.07, 0.27]	0.0023	0.0033
gpt-5.1	Careful	0.74	-0.22	[-0.31, -0.13]	<0.001	<0.001
gpt-5.4	Basic	0.76	-0.1	[-0.19, -0.01]	0.0639	0.0752
gpt-5.4	Simplified	0.71	0.13	[0.03, 0.23]	0.0241	0.0301
gpt-5.4	Practical	0.77	0.13	[0.04, 0.22]	0.0106	0.0142
gpt-5.4	Careful	0.69	-0.29	[-0.38, -0.20]	<0.001	<0.001
claude-sonnet-4-6	Basic	0.79	0.07	[-0.02, 0.16]	0.1892	0.1892
claude-sonnet-4-6	Simplified	0.69	0.19	[0.09, 0.29]	0.0009	0.0014
claude-sonnet-4-6	Practical	0.59	0.31	[0.20, 0.42]	<0.001	<0.001
claude-sonnet-4-6	Careful	0.75	-0.17	[-0.26, -0.08]	0.0009	0.0014
deepseek-chat-v3.2	Basic	0.72	0.2	[0.10, 0.30]	0.0002	0.0004
deepseek-chat-v3.2	Simplified	0.64	0.28	[0.18, 0.38]	<0.001	<0.001
deepseek-chat-v3.2	Practical	0.65	0.31	[0.21, 0.41]	<0.001	<0.001
deepseek-chat-v3.2	Careful	0.79	-0.19	[-0.27, -0.11]	<0.001	<0.001
GPT-4.1	Basic	0.75	0.21	[0.12, 0.30]	<0.001	<0.001
GPT-4.1	Careful	0.84	-0.08	[-0.16, 0.00]	0.0768	0.0809
GPT-4.1	Practical	0.64	0.32	[0.22, 0.42]	<0.001	<0.001
GPT-4.1	Simplified	0.7	0.26	[0.17, 0.35]	<0.001	<0.001

These values correspond to the 2×2 contingency tables presented in Supplementary Table 7. Exact McNemar tests were used to assess whether the rate of causal language in model-generated summaries differed from that in the paired abstracts for each model–prompt condition. We also report differences in proportions (Δ Proportion), with positive values indicating that summaries contained more causal language and negative values indicating that abstracts did, along with 95% Wald Confidence Intervals. p_{adj} denotes p-values after FDR adjustment for multiple comparisons. Note that each row in this table compares 100 summaries to 100 abstracts.

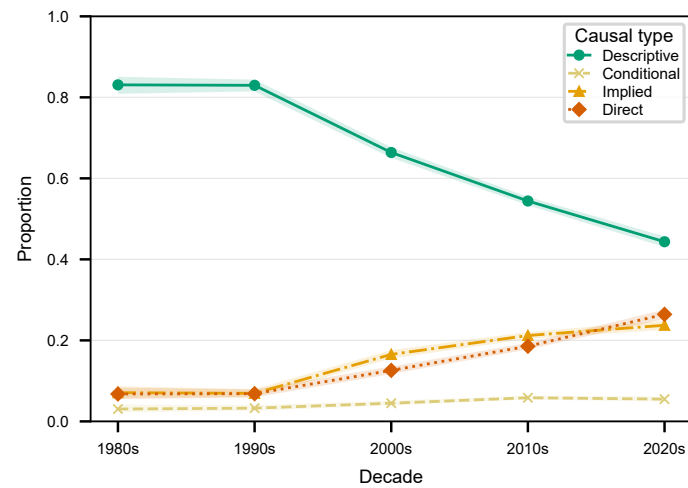


Extended Data Fig. 1 | Annual count of articles in our dataset and proportion labeled cross-sectional. The number of annual articles in our dataset (black) drastically increased over time. The proportion of articles identified by our

keywords and labeled as cross-sectional by our classifier (gray) increased until 1980 but has stayed relatively stagnant since. The sharp decline in articles for the last year (2025) is a product of when data were collected, the beginning of that year.



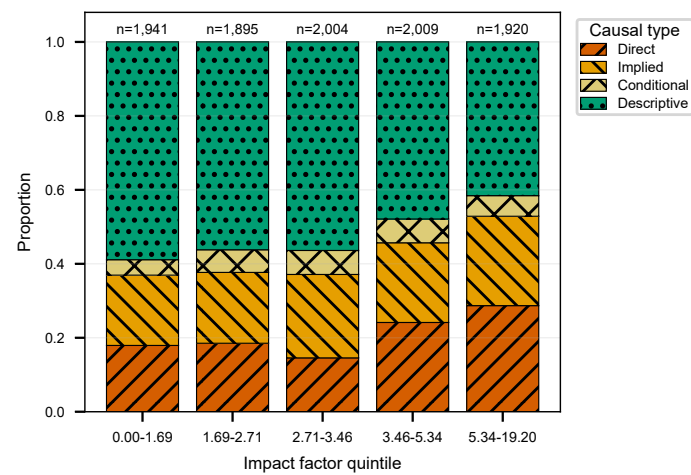
Extended Data Fig. 2 | Count of articles by discipline. A count of cross-sectional articles by discipline reveals that major social science disciplines had strong coverage within the dataset.



Extended Data Fig. 3 | Proportion of articles by causal type by decade.

Decade proportions of cross-sectional articles by claim type. Shaded portions display 95% Wilson CIs. For this figure, we relied on the smaller sample of article abstracts that we were able to export from TDM Studio with the following counts

by decade: N's 1980 = 1,212, 1990 = 2,572, 2000 = 4,947, 2010 = 8,910, 2020 = 5,441. The decline in purely descriptive language is driven primarily by increases in direct and implied causal statements. Refer to the Supplemental Information "Results by Causal Type" for additional details.



Extended Data Fig. 4 | Proportion of articles by causal type and journal impact factor quintile. Claim-type distribution by journal impact-factor quintile. Higher-impact journals exhibit more causal language overall, particularly direct and implied claims. Refer to the Supplemental Information “Results by Causal Type” for additional details.

a	b	c	d
Original	Rewrite	Methodological Note	“AI” Feedback
Title: The role of Independent Variable in shaping Dependent Variable Abstract: In this study, we ran a cross-sectional survey and found an influence of Independent Variable on Dependent Variable, matching theoretical expectations which anticipate such a relationship.	Title: The associations between Independent Variable and Dependent Variable Abstract: In this study, we ran a cross-sectional survey and found a correlation between Independent Variable and Dependent Variable, matching theoretical expectations which anticipate such a relationship. Future work is necessary to understand the causal factors underlying this relationship.	Title: The role of Independent Variable in shaping Dependent Variable Abstract: In this study, we ran a cross-sectional survey and found an influence of Independent Variable on Dependent Variable, matching theoretical expectations which anticipate such a relationship. Note: This study looks at data from one point in time, so it can only show patterns or correlations between things—it can't show that one thing causes another or predicts a future outcome.	Title: The role of Independent Variable in shaping Dependent Variable Abstract: In this study, we ran a cross-sectional survey and found an influence of Independent Variable on Dependent Variable, matching theoretical expectations which anticipate such a relationship. Your AI reviewer for accurate insights into academic texts. ACE This study looks at data from one point in time, so it can only show patterns or correlations between things—it can't show that one thing causes another or predicts a future outcome. ACE highlighted certain phrases that might give the impression of a causal link. Keep in mind: these associations do not imply cause and effect.

Extended Data Fig. 5 | Experimental conditions. Participants were randomly assigned to one of four conditions. a, “Original”: participants viewed the publication title and abstract verbatim. b, “Rewrite”: participants viewed a minimally edited version in which causal claims were replaced with associational language. c, “Methodological Note”: participants viewed the original title and abstract accompanied by a brief note describing the study design and its

inferential limitations. d, “AI Feedback”: participants viewed the original title and abstract with highlighted instances of overclaiming and an accompanying note, both presented as AI-generated feedback. These examples contain artificial, illustrative titles and abstracts, but the stimuli used in the experiment come from actual peer-reviewed publications.

EXTENDED DATA BOX 1

GPT-4o Prompt to Detect Cross-sectional Methodology

Read the academic work and determine the study design. Classify the Study: Assign the paper to exactly one of the following categories based on its methodology:

- Cross-sectional - Data collected at a single time point only, typically via a one-time survey or a single snapshot from a database.
 - Mixed Methods - Both quantitative (e.g., surveys) and qualitative (e.g., interviews, focus groups, textual coding) are used in parallel.
 - Qualitative - Relies solely on interviews, focus groups, narrative data, thematic analyses, or phenomenological approaches.
 - Randomized Control Trial (RCT) - Participants are randomly assigned to intervention/control conditions.
 - Quasi-experimental - Involves an intervention, but no full random assignment.
 - Multi-Time Point - Longitudinal, data collected across multiple waves/time points (panel or repeated measures).
 - Review/Meta-analysis - Summarizes, critiques, or synthesizes findings from multiple other studies; no new primary data analysis.
 - Multiple Studies - The paper explicitly includes more than one independent empirical study.
 - Other - For example, secondary commentary, editorials, legal analyses, policy briefs, conceptual/theoretical articles, or anything that does not fit the above categories.
- Important: 1. Classify a study as Cross-sectional ONLY if the paper explicitly states that it relies on data from one time point or one wave without any mention of multiple years, interventions, or additional analyses. 2. If the methodology is ambiguous, err on the side of not choosing Cross-sectional (to avoid false positives). 3. Return only one word from the category list above (e.g., Cross-sectional, Mixed, Qualitative, RCT, Quasi-experimental, Multi-Time, Review, Multiple, or Other).

EXTENDED DATA BOX 2

GPT-4o Prompt to Detect Causal Language

Causal language can appear in abstracts in different forms:

- 1Direct causal statement - The authors explicitly claim one variable affects another in their results or conclusions (e.g., “influences,” “causes,” “impacts,” “mechanisms”).
- 2Implied causal introduction - The authors state they are investigating a causal relationship in the introduction, without direct causal claims in the results (e.g., “this study examines the influence”).
- 3Conditional causal language - The authors make causal claims but hedge their language (e.g., “may cause,” “suggests a role of”).
- 4Implied causal conclusion - The

authors imply causality in their conclusions, such as recommending interventions based on their findings.

- 5None - The authors use purely associational language when discussing their findings.

Instructions: Code the presence of causal language in this abstract based on the categories above. Ignore causal language referring to prior studies or general background. Focus only on how the authors describe their own, novel, empirical results. Respond with a single number. If multiple types are present, report the lowest applicable number.

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	DESCRIPTIVE ANALYSIS: Article labels were generated using OpenAI GPT models accessed through the OpenAI API. BERT-based classifiers were trained and applied in Python. Article information was accessed on ProQuest's TDM Studio platform. EXPERIMENT: The human-subjects experiment was administered in Qualtrics. OpenAI GPT models were used to classify selected participant responses and generate AI-based summaries. Platform API's were also used for Claude and Deepseek Models. Statistical analyses were conducted in R and Python.
Data analysis	All analysis was conducted in Python and R. Complete reproduction materials are publicly available online: https://osf.io/zpua6/. This folder contains scripts, datasets, and package information for reproduction of the results, though the complete text for all articles in our descriptive analysis must be accessed on TDM Studio. Package versions for python and R are listed below, and the OSF repository provides instructions for how to install these (along with all their appropriate dependencies, and additional details for supplemental analysis) to enable full reproduction. Python: Python 3.9.6, pandas 2.3.3, NumPy 2.0.2, SciPy 1.13.1, statsmodels 0.14.6, scikit-learn 1.6.1, matplotlib 3.9.4, openpyxl 3.1.5 R: R 4.5.1, lme4 1.1-37, lmerTest 3.1-3, dplyr 1.1.4, broom 1.0.9, emmeans 1.11.2, ggplot2 3.5.2, clubSandwich 0.6.1

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All article-level (including citation information) and experimental results datasets required to reproduce the findings of this study are available on the Open Science Framework (OSF) at: <https://osf.io/zpua6/>. The underlying article texts and metadata were obtained through ProQuest TDM Studio and are subject to licensing restrictions. These data are therefore not redistributed by the authors but may be obtained directly through ProQuest's TDM Studio platform (<https://tdmstudio.proquest.com/home>) subject to institutional access and licensing agreements.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	We do not report on sex or gender differences in this study.
Reporting on race, ethnicity, or other socially relevant groupings	We do not report on race, ethnicity, or other socially relevant differences in this study.
Population characteristics	See above
Recruitment	We recruited 1,100 adult participants through Prolific using the Wharton Behavioral Lab. Eligibility was restricted to individuals who hold at least a bachelor's degree and reside in the United States. Participants received fair monetary compensation consistent with Prolific's recommended hourly rate.
Ethics oversight	The study protocol has been reviewed and approved by the University of Pennsylvania's Institutional Review Board (IRB #858559). Participants provided informed consent electronically and could withdraw at any time. Data were collected anonymously and stored in compliance with IRB guidelines.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This paper contains two quantitative analyses: A large scale descriptive study on the presence of causal claims in cross sectional studies and an experiment that assesses how these claims impact reader (and LLM) interpretations of a studies implications.
Research sample	DESCRIPTIVE ANALYSIS: The final dataset contained 194,631 cross-sectional articles published between 1980 and 2024. Articles were drawn from five ProQuest TDM Studio databases and included article titles, abstracts, and associated metadata. Duplicate records with different identifiers but identical titles were removed. EXPERIMENT: The experiment included 1,105 participants recruited through Prolific by the Wharton Behavioral Lab. Eligibility was restricted to adults residing in the United States who held at least a bachelor's degree. The target sample size was 1,100 participants; Prolific slightly over-recruited, resulting in a final sample of 1,105 participants. No demographic information beyond these eligibility criteria was collected. Because participation on Prolific is voluntary, the sample reflects a self-selected group of online research participants and is not intended to be representative of the broader U.S. population. LLM EXPERIMENTS: The language-model experiments examined models from three leading AI providers at the time of data collection: OpenAI (GPT-4.1, GPT-5.1, and GPT-5.4), Anthropic (Claude Sonnet 4.6), and DeepSeek (DeepSeek Chat-V3.2).
Sampling strategy	DESCRIPTIVE ANALYSIS: Articles were identified using a keyword-based search of five ProQuest TDM Studio databases and therefore constitute a purposive rather than random sample of likely cross-sectional studies. The resulting corpus was intended to provide broad coverage of cross-sectional social-science research rather than a representative sample of all academic publications. No formal

sample size calculation was performed. Instead, all eligible articles returned by the search and classification procedures were retained, resulting in a final sample of 194,631 cross-sectional articles.

EXPERIMENT: Participants were recruited through Prolific using a convenience sample restricted to U.S. adults with at least a bachelor's degree. A priori power analyses were conducted (full details in main text) for establishing our target sample. The final sample size of 1,105 participants slightly exceeded this target. Although not nationally representative, the sample was selected to provide a sufficiently powered test of the experimental manipulations.

LLM EXPERIMENTS: Models were purposively selected from three leading AI providers and were not intended to represent all available language models. No formal power analyses were conducted. The first GPT-4.1 experiment was exploratory. For the second LLM experiment, 100 articles were selected to provide a diverse set of stimuli spanning multiple causal-language categories while remaining computationally feasible. Findings should therefore be interpreted as evidence regarding the sampled models and articles rather than all possible language models or research papers.

Data collection

DESCRIPTIVE ANALYSIS: Article titles, abstracts, and metadata were collected through ProQuest TDM Studio. A random sample of 40,000 articles was exported and labeled using GPT-4o via the OpenAI API. These labels were used to train local BERT-based classifiers, which were subsequently applied to the full corpus within the secure TDM Studio environment. The resulting article-level labels, together with citation information and unique identifiers, were exported for statistical analysis and are publicly available. No human participants were involved, and researcher blinding was therefore not applicable.

EXPERIMENT: Data were collected using Qualtrics and administered online through Prolific. Participants completed the study remotely without the presence of researchers or other participants. Assignment to experimental condition was performed automatically through Qualtrics using embedded JavaScript randomization procedures. Researchers were aware of the study hypotheses but did not interact directly with participants during data collection. Because assignment was randomized and all procedures, exclusion criteria, and analyses were preregistered prior to data collection, researcher awareness of the hypotheses could not influence condition assignment or participant responses. Raw Qualtrics data were exported following data collection and all cleaning and analysis procedures are documented in the public OSF repository.

LLM EXPERIMENTS: Data were collected through the application programming interfaces (APIs) of OpenAI, Anthropic, and DeepSeek. Models were queried programmatically using standardized prompts and parameter settings described in the manuscript and Supplementary Information. Responses were recorded automatically and stored for analysis. No human participants were involved, and no researchers were present during data generation. Researcher blinding was not applicable because the outputs were generated through automated API calls using predefined procedures.

Timing

DESCRIPTIVE ANALYSIS: The ProQuest TDM Studio databases were queried on February 7, 2025. A training dataset of 30,000 articles was labeled using GPT-4o through OpenAI's Batch API on March 2, 2025. A separate validation dataset of 10,000 articles was labeled using the same procedure on April 29, 2025. The resulting classifiers were deployed within TDM Studio and applied to the full corpus during March and April 2025.

EXPERIMENT: The GPT-4.1 experiment was conducted using OpenAI's Batch API on July 3, 2025. The human-subjects experiment was conducted on Prolific on July 14, 2025.

SECOND LLM EXPERIMENT: The second LLM experiment, which evaluated multiple language models, prompting strategies, and article samples, was conducted between March 25 and March 28, 2026.

Data exclusions

DESCRIPTIVE ANALYSIS: Duplicate articles with identical titles but multiple identifiers were excluded prior to analysis (N = 6,276, 1.9%). Analyses reported in the main text were additionally restricted to articles published in peer-reviewed academic journals; non-journal publications (e.g., dissertations, conference proceedings, reports) were excluded from the primary analyses (N = 318,357). Both exclusion criteria were established prior to analysis. Supplementary analyses including all publication types yielded substantively similar results.

EXPERIMENT: No participants were excluded from the primary analyses. A preregistered attention check was included, and supplementary sensitivity analyses excluding participants who either failed this check, or completed the task in less than 1/3 the median pilot time (N = 36) are reported in the Supplementary Information.

LLM EXPERIMENTS: No model-generated responses or summaries were excluded from the analyses. All successfully generated outputs were retained and analyzed according to the preregistered procedures.

Non-participation

No participants dropped out or declined participation.

Randomization

DESCRIPTIVE ANALYSIS: No randomization was performed because this component of the study analyzed an existing corpus of published articles.

EXPERIMENT: Participants were randomly assigned by Qualtrics to one of five conditions: Original (N = 216), Rewrite (N = 240), Methodological Note (N = 229), AI Feedback (N = 202), and No Information (N = 218). Randomization was performed automatically by the survey software prior to participants viewing study materials. Additional details regarding the experimental conditions are provided in the manuscript.

LLM EXPERIMENTS: No randomization was performed. For each article and prompt condition, responses were generated separately from each language model through independent API calls. All models received the same stimuli and prompts, allowing direct comparisons across models and prompting conditions.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involvement in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA